FOUNDATIONONE®CDx

PATIENT	TUMOR TYPE	REPORT DATE
Tang, Mei Hui	Brain glioblastoma (GBM)	31 May 2023
	COUNTRY CODE	ORDERED TEST #
	TW	ORD-1634763-01

ABOUT THE TEST FoundationOne®CDx is a next-generation sequencing (NGS) based assay that identifies genomic findings within hundreds of cancer-related genes.

PATIENT
DISEASE Brain glioblastoma (GBM)
NAME Tang, Mei Hui
DATE OF BIRTH 12 November 1968
SEX Female
MEDICAL RECORD # 48396970

PHYSICIAN
ORDERING PHYSICIAN Yeh, Yi-Chen
MEDICAL FACILITY Taipei Veterans General Hospital
ADDITIONAL RECIPIENT None
MEDICAL FACILITY ID 205872
PATHOLOGIST Not Provided

SPECIMEN
SPECIMEN SITE Brain
SPECIMEN ID S112-19517A (PF23059)
SPECIMEN TYPE Slide Deck
DATE OF COLLECTION 02 May 2023
SPECIMEN RECEIVED 22 May 2023

Biomarker Findings

Microsatellite status - MS-Stable
Tumor Mutational Burden - 10 Muts/Mb

Genomic Findings

For a complete list of the genes assayed, please refer to the Appendix.

EGFR amplification
KIT amplification
PDGFRA amplification
ARID1A E1924*
CDK4 amplification
MDM2 amplification
MYC amplification
PIK3R1 K80fs*26
KDM5C inversion exons 16-19
NOTCH1 E1595fs*13 - subclonal†
TERT promoter -146C>T

1 Disease relevant genes with no reportable alterations: ***IDH1***

† See About the Test in appendix for details.

Report Highlights

- Variants with **diagnostic implications** that may indicate a specific cancer type: ***EGFR* amplification** (p. 4), ***TERT* promoter -146C>T** (p. 10)
- Targeted therapies with potential clinical benefit **approved in another tumor type**: **Cetuximab** (p. 11), **Imatinib** (p. 11), **Nilotinib** (p. 12), **Panitumumab** (p. 12), **Sorafenib** (p. 13), **Sunitinib** (p. 13)
- Evidence-matched **clinical trial options** based on this patient's genomic findings: (p. 14)
- Variants with **prognostic implications** for this tumor type that may impact treatment decisions: ***TERT* promoter -146C>T** (p. 10)

BIOMARKER FINDINGS	THERAPY AND CLINICAL TRIAL IMPLICATIONS
Microsatellite status - MS-Stable	**No therapies or clinical trials.** See Biomarker Findings section
Tumor Mutational Burden - 10 Muts/Mb	**No therapies or clinical trials.** See Biomarker Findings section

GENOMIC FINDINGS	THERAPIES WITH CLINICAL RELEVANCE (IN PATIENT'S TUMOR TYPE)	THERAPIES WITH CLINICAL RELEVANCE (IN OTHER TUMOR TYPE)
EGFR - amplification	none	Cetuximab
4 Trials see p. 18		Panitumumab
KIT - amplification	none	Imatinib
		Nilotinib
		Sorafenib
10 Trials see p. 19		Sunitinib

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENOMIC FINDINGS	THERAPIES WITH CLINICAL RELEVANCE (IN PATIENT'S TUMOR TYPE)	THERAPIES WITH CLINICAL RELEVANCE (IN OTHER TUMOR TYPE)
PDGFRA - amplification	none	Imatinib
2 Trials see p. 24		
ARID1A - E1924*	none	none
6 Trials see p. 14		
CDK4 - amplification	none	none
10 Trials see p. 16		
MDM2 - amplification	none	none
5 Trials see p. 21		
MYC - amplification	none	none
8 Trials see p. 22		
PIK3R1 - K80fs*26	none	none
3 Trials see p. 25		

GENOMIC FINDINGS WITH NO REPORTABLE THERAPEUTIC OR CLINICAL TRIAL OPTIONS

For more information regarding biological and clinical significance, including prognostic, diagnostic, germline, and potential chemosensitivity implications, see the Genomic Findings section.

KDM5C - inversion exons 16-19 ..... p. 9

NOTCH1 - E1595fs*13 - subclonal ..... p. 9

TERT - promoter -146C>T ..... p. 10

NOTE Genomic alterations detected may be associated with activity of certain approved therapies; however, the agents listed in this report may have varied clinical evidence in the patient's tumor type. Therapies and the clinical trials listed in this report may not be complete and exhaustive. Neither the therapeutic agents nor the trials identified are ranked in order of potential or predicted efficacy for this patient, nor are they ranked in order of level of evidence for this patient's tumor type. This report should be regarded and used as a supplementary source of information and not as the single basis for the making of a therapy decision. All treatment decisions remain the full and final responsibility of the treating physician and physicians should refer to approved prescribing information for all therapies.

Therapies contained in this report may have been approved by the US FDA.

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

BIOMARKER

Microsatellite status

RESULT
MS-Stable

POTENTIAL TREATMENT STRATEGIES

— Targeted Therapies —

On the basis of clinical evidence, MSS tumors are significantly less likely than MSI-H tumors to respond to anti-PD-1 immune checkpoint inhibitors[1-3], including approved therapies nivolumab and pembrolizumab[4]. In a retrospective analysis of 361 patients with solid tumors treated with pembrolizumab, 3% were MSI-H and experienced a significantly higher ORR compared with non-MSI-H cases (70% vs. 12%, p=0.001)[5].

FREQUENCY & PROGNOSIS

Low-level MSI has been reported in 5-9% of glioblastoma (GBM) samples[6-8]. A large-scale study did not find high-level microsatellite instability (MSI-H) in any of 129 GBM samples[6], although a small-scale study reported MSI-H in 4 of 15 pediatric GBMs and 1 of 12 adult GBMs[9]. The frequency of MSI has been reported to be increased in relapsed compared to primary GBM[6], in GBMs with a previous lower grade astrocytoma[7], and in giant cell GBM compared to classic GBM[8].

FINDING SUMMARY

Microsatellite instability (MSI) is a condition of genetic hypermutability that generates excessive amounts of short insertion/deletion mutations in the genome; it generally occurs at microsatellite DNA sequences and is caused by a deficiency in DNA mismatch repair (MMR) in the tumor[10]. Defective MMR and consequent MSI occur as a result of genetic or epigenetic inactivation of one of the MMR pathway proteins, primarily MLH1, MSH2, MSH6, or PMS2[10-12]. This sample is microsatellite-stable (MSS), equivalent to the clinical definition of an MSS tumor: one with mutations in none of the tested microsatellite markers[13-15]. MSS status indicates MMR proficiency and typically correlates with intact expression of all MMR family proteins[10,12,14-15].

BIOMARKER

Tumor Mutational Burden

RESULT
10 Muts/Mb

POTENTIAL TREATMENT STRATEGIES

— Targeted Therapies —

On the basis of clinical evidence in solid tumors, increased TMB may be associated with greater sensitivity to immunotherapeutic agents, including anti-PD-L1[16-18], anti-PD-1 therapies[16-19], and combination nivolumab and ipilimumab[20-25]. In glioma, a lack of association between TMB and clinical benefit from immune checkpoint inhibitors has been reported[16,26-27]. However, multiple case studies have reported that patients with ultramutated gliomas driven by POLE mutations have benefited from treatment with anti-PD-1[28-29] or anti-PD-L1[30] therapies. Therefore, although increased TMB alone may not be a strong biomarker for PD-1 or PD-L1 inhibitors in this cancer type, these agents may have efficacy for patients with glioma harboring both high TMB and POLE mutation.

FREQUENCY & PROGNOSIS

Glioblastoma (GBM) harbors a median TMB of 2.7 mutations per megabase (muts/Mb), and 4.2% of cases have high TMB (>20 muts/Mb)[31]. For pediatric patients, high TMB has been reported in a subset of high-grade gliomas, frequently in association with mutations in mismatch repair or proofreading genes and in TP53, whereas BRAF alterations or other oncogene fusions were observed more frequently in brain tumors harboring low TMB[32-33]. Increased TMB has been reported to correlate with higher tumor grade in glioma[34] and glioblastoma (GBM) tissue samples with biallelic mismatch repair deficiency (bMMRD)[28], as well as with shorter OS of patients with diffuse glioma[35].

FINDING SUMMARY

Tumor mutation burden (TMB, also known as mutation load) is a measure of the number of somatic protein-coding base substitution and insertion/deletion mutations occurring in a tumor specimen. TMB is affected by a variety of causes, including exposure to mutagens such as ultraviolet light in melanoma[36-37] and cigarette smoke in lung cancer[38-39], treatment with temozolomide-based chemotherapy in glioma[40-41], mutations in the proofreading domains of DNA polymerases encoded by the POLE and POLD1 genes[42-46], and microsatellite instability (MSI)[42,45-46]. This sample harbors a TMB below levels that would be predicted to be associated with sensitivity to PD-1- or PD-L1-targeting immune checkpoint inhibitors, alone or in combination with other agents, for patients with glioma[16,26-30]. Although efficacy of immune checkpoint inhibitors has been observed for patients with other solid tumor types harboring TMB levels such as seen here[16-19,47], an association between TMB and clinical benefit has generally not been observed for patients with glioma[16,26-27], except for those with ultramutated glioma with POLE mutation[28-30].

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE
EGFR

ALTERATION
amplification

POTENTIAL TREATMENT STRATEGIES

— Targeted Therapies —

Clinical studies of the second-generation EGFR TKIs afatinib and dacomitinib for patients with EGFR-amplified gliomas have shown limited efficacy[48-52]; however, a small subset of patients has experienced clinical benefit[48-50]. Multiple studies have failed to find a positive association between increased EGFR expression and clinical benefit from erlotinib or gefitinib for patients with glioblastoma[53-56]. There are conflicting data on the efficacy of anti-EGFR antibodies for the treatment of EGFR-amplified tumors. A meta-analysis of colorectal cancer patients treated with second-line or higher cetuximab or panitumumab observed an association between EGFR copy number gain and increased OS and PFS[57]. However, studies in head and neck squamous cell carcinoma and gastric cancer found either no association or a negative association between EGFR copy number gain and survival after treatment with first-line cetuximab or panitumumab in combination with chemotherapy[58-59]. The Phase 3 INTELLANCE trial of depatuxizumab mafodotin (ABT-414), an EGFR-targeted antibody-drug conjugate with a toxic payload, in patients with EGFR-amplified glioblastoma (GBM) was stopped for futility. Interim analysis demonstrated improved median PFS (mPFS) of ABT-414 monotherapy compared with placebo (HR=0.84); however, no OS benefit was observed (HR=1.01). Improved mPFS was also observed in patients harboring EGFRvIII (HR=0.73) but without an OS improvement (HR=0.95)[60]. The Phase 2 INTELLANCE trial demonstrated clinical benefit for EGFR-amplified GBM for the combination of ABT-414, temozolomide, and radiotherapy (HR=0.66, p=0.017), but there was no evidence of efficacy for ABT-414 monotherapy (HR=1.04, p=0.83)[61]. A single-center Phase 2 pilot study for EGFR-amplified solid tumors (n=13) reported 1 PR and 2 SDs, including 1 with a 9-month duration, for the 8 evaluable patients with glioblastoma[62].

FREQUENCY & PROGNOSIS

Across several genomic studies of CNS tumors, EGFR amplification has been reported in 16.9% of anaplastic astrocytomas, and 39.7% of glioblastoma multiformes (GBMs)[63-66]. In a study of IDH-wildtype GBM samples, EGFR alterations were detected in 50% (117/232) of IDH-wildtype GBM samples analyzed, including 41% (95/232) with a co-occurring EGFR amplification and mutation, 26% (61/232) with an EGFR domain truncation event, such as EGFRvIII, and 2.2% (5/232) with an EGFR fusion event[67]. No definitive correlation has been identified between EGFR amplification and length of survival in patients with GBM[68-69]; however, EGFR amplification has been associated with prolonged survival in patients over the age of 60 with GBM[70].

FINDING SUMMARY

EGFR encodes the epidermal growth factor receptor, which belongs to a class of proteins called receptor tyrosine kinases. In response to signals from the environment, EGFR passes biochemical messages to the cell that stimulate it to grow and divide[71]. Amplification of EGFR has been associated with increased expression of EGFR mRNA and protein in several cancer types[72-74].

POTENTIAL DIAGNOSTIC IMPLICATIONS

The presence of EGFR gene amplification or TERT promoter mutations are indicative of diffuse astrocytic glioma with molecular features of glioblastoma, WHO grade 4 in IDH1/2-wildtype tumors (NCCN CNS Cancers Guidelines, v2.2022)[75].

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE
KIT

ALTERATION
amplification

POTENTIAL TREATMENT STRATEGIES

— Targeted Therapies —

On the basis of clinical evidence, primarily in gastrointestinal stromal tumor (GIST), melanoma, AML, and systemic mastocytosis, KIT activating alterations are associated with sensitivity to TKIs including imatinib, sunitinib, sorafenib, dasatinib, nilotinib, pazopanib, regorafenib, ponatinib, midostaurin, apatinib, avapritinib, and ripretinib[76-85]. The use of mTOR inhibitors as an alternative therapeutic strategy has demonstrated limited success in KIT-mutated, imatinib-resistant melanoma, with 1 PR and 3 SD observed for 4 patients treated with everolimus[86]. However, no responses were observed for 10 patients with mastocytosis following everolimus monotherapy, with 8/10 patients harboring the KIT D816V mutation[87]. The role of KIT amplification as a biomarker for response to mTOR inhibitors has not been investigated (PubMed, Feb 2023). Clinical benefit has been observed for patients with KIT amplified or overexpressing tumors following treatment with imatinib[88-98], nilotinib[99], sorafenib[100-103], and sunitinib[104-105], suggesting that KIT amplification may be sensitive to these inhibitors. However, evidence demonstrating clinical benefit for regorafenib, dasatinib, pazopanib, or ponatinib in the context of KIT amplified or overexpressing tumors is limited.

FREQUENCY & PROGNOSIS

In the TCGA datasets, KIT amplification has been reported in 2.5% of lower grade gliomas (grades 2 and 3)[106] and 9.2% of glioblastomas (Grade 4 astrocytoma)[64]. KIT amplification has been variously reported in 4-47% of glioblastomas in the scientific literature[107-109]. Amplification of KIT has been strongly correlated with the presence of KDR and/or PDGFRA amplification in glioblastoma[108,110-111]. One study found no correlation between KIT amplification and OS for patients with glioblastoma, while a separate study reported that overexpression of KIT was associated with tumor grade and shorter survival for patients with malignant glioma[107,112].

FINDING SUMMARY

KIT (also called c-KIT) encodes a cell surface tyrosine kinase receptor that, upon ligand binding and dimerization, activates the PI3K-AKT and RAS-MAPK signaling pathways[113]. KIT aberrations, including point mutations, translocations, amplification, and overexpression, have been associated with various malignancies, and KIT is considered an oncoprotein[114]. KIT has been reported to be amplified in cancer[115] and may be biologically relevant in this context[116-117].

GENE
PDGFRA

ALTERATION
amplification

POTENTIAL TREATMENT STRATEGIES

— Targeted Therapies —

On the basis of extensive clinical evidence in solid tumors and hematologic cancers, PDGFRA activating alterations are associated with sensitivity to imatinib[118-155]. Sorafenib has shown clinical and preclinical activity against the FIP1L1-PDGFRA fusion in chronic eosinophilic leukemia (CEL) and mutations associated with clinical resistance to imatinib and sunitinib in both CEL and gastrointestinal stromal tumor (GIST)[156-161]. Complete responses to nilotinib have been reported in patients with CEL or hypereosinophilic syndrome with FIP1L1-PDGFRA or activating mutations[134,162-163]; preclinical evidence has reported efficacy of nilotinib in the context of PDGFRA mutations associated with GIST[164-165]. Patients with GIST harboring PDGFRA activating mutations have been reported to derive clinical benefit from treatment with sunitinib[166] or regorafenib[167-168]. Preclinical studies have reported sensitivity of activating PDGFRA mutations and FIP1L1-PDGFRA fusion to dasatinib[158,164]. PDGFRA D842 mutations were reported to be sensitive to avapritinib in clinical[76] and preclinical[76] studies of GIST, and demonstrated sensitivity to ripretinib for 1 patient[169].

FREQUENCY & PROGNOSIS

PDGFRA amplification has been suggested to be more common in higher grade astrocytomas than in lower grade astrocytomas; studies have reported PDGFRA amplification in 16.3% (27/166) of Grade 2 astrocytomas and in 23.6% (91/386) of Grade 3 and 4 astrocytomas analyzed[110,170-171]. PDGFRA amplification has been reported in 5.2-33% of glioblastoma cases[64,107-108,170,172-173]. A retrospective analysis of TCGA glioma samples reported elevated expression of ERBB3 correlated with PDGFRA expression and co-expression of these genes was an indicator of poor prognosis in a GBM patient cohort[174]. PDGFRA amplification has been associated with tumor grade and poor PFS and OS for patients with glioblastoma[170,172-173]. In addition, PDGFRA amplification has been reported to occur in conjunction with IDH1 mutations in glioblastoma, and both alterations in the same tumor have been associated with poor patient prognosis[170]. Amplification of PDGFRA has also been strongly correlated with the presence of KDR and/or KIT amplification in glioblastomas, as well as with EGFR amplification[108,110-111,175].

FINDING SUMMARY

PDGFRA encodes platelet-derived growth factor receptor alpha (PDGFR-alpha), a tyrosine kinase receptor that, upon binding of cognate ligands (PDGFA or PDGFB), activates several signaling pathways, including PI3K and MAPK[176]. PDGFR aberrations, including point mutations, translocations, amplification, and/or overexpression, have been associated with various malignancies[114]. Amplification of PDGFRA, frequently occurring with amplification of the genes KDR and KIT, has been associated with increased PDGFRA expression[109,177-179] and poor prognosis[109,170,180-181] in some subtypes of glioma.

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE

ARID1A

ALTERATION
E1924*

HGVS VARIANT
NM_006015.4:c.5770G>T (p.E1924*)

VARIANT CHROMOSOMAL POSITION
chr1:27106159

VARIANT ALLELE FREQUENCY (% VAF)
36.2%

POTENTIAL TREATMENT STRATEGIES

— Targeted Therapies —

There are no therapies approved to address the mutation or loss of ARID1A in cancer. However, on the basis of limited clinical and preclinical evidence, ARID1A inactivating mutations may lead to sensitivity to ATR inhibitors such as M6620 and ceralasertib[182]. In a Phase 2 study of ceralasertib in solid tumors, 2 patients with endometrial carcinoma in the cohort with loss of ARID1A expression achieved CRs on ceralasertib monotherapy; at least 1 of these 2 patients carried an inactivating ARID1A mutation. In contrast, no responses were observed for patients with normal ARID1A expression treated with ceralasertib combined with olaparib[183]. One patient with small cell lung cancer harboring an ARID1A mutation experienced a PR when treated with M6620 combined with topotecan[184]. In a Phase 1 trial, a patient with metastatic colorectal cancer (CRC) harboring both an ARID1A mutation and ATM loss treated with single-agent M6620 achieved a CR that was ongoing at 29 months[185]. On the basis of limited clinical and preclinical evidence, ARID1A inactivation may predict sensitivity to EZH2 inhibitors[186-187]. A Phase 1 study of EZH2 inhibitor CPI-0209 reported 1 PR for a patient with ARID1A-mutated endometrial cancer[188]. Other studies have reported that the loss of ARID1A may activate the PI3K-AKT pathway and be linked with sensitivity to inhibitors of this pathway[189-191]. Patients with ARID1A alterations in advanced or metastatic solid tumors may derive benefit from treatment with anti-PD-1 or anti-PD-L1 immunotherapy[192]. Loss of ARID1A expression has been associated with chemoresistance to platinum-based therapy for patients with ovarian clear cell carcinoma[193-194] and to 5-fluorouracil in CRC cell lines[195].

FREQUENCY & PROGNOSIS

ARID1A alterations are particularly prevalent in ovarian clear cell carcinoma (46-50%), ovarian and uterine endometrioid carcinomas (24-44%), and cholangiocarcinoma (27%); they are also reported in up to 27% of gastric carcinoma, esophageal adenocarcinoma, Waldenstrom macroglobulinemia, pediatric Burkitt lymphoma, hepatocellular carcinoma, colorectal carcinoma, and urothelial carcinoma samples analyzed (COSMIC, cBioPortal, 2023)[115,196-203]. ARID1A loss is associated with microsatellite instability in ovarian and endometrial endometrioid adenocarcinomas[192,204-207], CRC[192,208-210], and gastric cancer[192,211-215]. ARID1A protein loss is associated with tumors of poor histological grade for many tumor types, including colorectal cancer (CRC)[208-210], cervical cancer[216-217], gastric cancer[211-215], urothelial carcinoma[218-220], ovarian and endometrial cancers[194,204-207,221-225], breast carcinoma[226-228], and clear cell renal cell carcinoma[229]; ARID1A mutation has been associated with poor outcomes for patients with cholangiocarcinoma[230-233]. However, prognostic data regarding patient survival are often mixed and conflicting.

FINDING SUMMARY

ARID1A encodes the AT-rich interactive domain-containing protein 1A, also known as Baf250a, a member of the SWI/SNF chromatin remodeling complex. Mutation, loss, or inactivation of ARID1A has been reported in many cancers, and the gene is considered a tumor suppressor[199,214,227,234-239]. ARID1A mutations, which are mostly truncating, have been identified along the entire gene and often correlate with ARID1A protein loss[199,212,235-236,240], whereas ARID1A missense mutations are mostly uncharacterized.

GENE

CDK4

ALTERATION
amplification

POTENTIAL TREATMENT STRATEGIES

— Targeted Therapies —

CDK4 amplification or activation may predict sensitivity to CDK4/6 inhibitors such as abemaciclib, palbociclib, and ribociclib[241-244]. Clinical benefit has been reported for limited tumor types including patients with CDK4-amplified liposarcoma and sarcoma in response to treatment with abemaciclib[245], palbociclib[241,246], and ribociclib[247].

FREQUENCY & PROGNOSIS

CDK4 amplification has been observed in 9.4% of glioma cases[63]. A study has reported amplification of the 12q14-15 region, where CDK4 and MDM2 reside, in 4.8% (2/42) of glioblastomas[248]. Amplification of CDK4 and corresponding increased CDK4 protein expression has been reported to be associated with a poorer patient outcome in anaplastic astrocytoma and glioblastoma[249-252].

FINDING SUMMARY

CDK4 encodes the cyclin-dependent kinase 4, which regulates the cell cycle, senescence, and apoptosis[253]. CDK4 and its functional homolog CDK6 are activated by D-type cyclins and promote cell cycle progression by inactivating the tumor suppressor Rb[254-255]. Amplification of the chromosomal region that includes CDK4 has been reported in multiple cancer types, including lung cancer, glioblastoma, and liposarcoma, and has been associated with overexpression of CDK4 protein[241,256-262].

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE

MDM2

ALTERATION
amplification

POTENTIAL TREATMENT STRATEGIES

— Targeted Therapies —

MDM2 antagonists disrupt the MDM2-p53 interaction, thereby stabilizing p53[263]. Preclinical studies have suggested that the amplification of MDM2, in the absence of concurrent TP53 mutations, may increase sensitivity to these agents[264-265]. Preliminary Phase 1 studies of the MDM2-p53 antagonist alrizomadlin (APG-115) reported a PR in a patient with liposarcoma harboring an MDM2 amplification and wildtype for TP53 and SD in 21%–38% (6/28 and 5/13, respectively) of patients in genomically unselected solid tumors[266-267] . A Phase 2 trial of alrizomadlin in combination with pembrolizumab reported a PR in 1 of 3 patients with malignant peripheral nerve sheath tumor that had failed standard therapy, as well as PRs in patients with multiple types of solid tumors that had failed immunotherapy, including 1 out of 14 patients with non-small cell lung cancer; 1 out of 5 patients with urothelial carcinoma; and 2 out of5, 1 out of 5, and 1 out of 11 patients with mucosal, uveal, and cutaneous melanoma, respectively[268]. Phase 1b studies of the MDM2 inhibitor idasanutlin for refractory AML in combination with cytarabine or venetoclax reported anti-leukemic response rates of 33% (25/75) and 37% (11/30), respectively[269-270] ; clinical benefit (58% ORR, 7/12) with idasanutlin monotherapy has been reported for patients with polycythemia vera[271]. The dual MDM2/MDM4 inhibitor ALRN-6924 led to an ORR of 27% (4/15) for patients with TP53 wildtype peripheral T-cell lymphoma in a Phase 2 study[272]; responses have also been observed in TP53 wildtype AML, MDS, Merkel cell carcinoma, colorectal cancer, and liposarcoma[273-274] .

FREQUENCY & PROGNOSIS

In the Glioblastoma Multiforme (GBM) TCGA dataset, amplification of MDM2 has been found in 8% of cases[64]. A study has reported amplification of the 12q14-15 region, where MDM2 and CDK4 reside, in 5% (2/42) of GBMs[248]. Amplification of MDM2 has been associated with poor survival in patients with glioblastoma[248,275]. Published data investigating the prognostic implications of MDM2 alterations in central nervous system malignancies are limited and conflicting. Multivariate analysis from one study suggests that MDM2 expression was associated with longer disease-free survival, while other studies reported that patients with MDM2-positive astrocytoma or glioblastoma multiforme experienced shorter overall survival[276-277].

FINDING SUMMARY

MDM2 encodes an E3 ubiquitin protein ligase, which mediates the ubiquitination and subsequent degradation of p53, Rb1, and other proteins[278-280]. MDM2 acts to prevent the activity of the tumor suppressor p53; therefore, overexpression or amplification of MDM2 may be oncogenic[281-282]. Overexpression or amplification of MDM2 is frequent in cancer[117]. Although two retrospective clinical studies suggest that MDM2 amplification may predict a short time-to-treatment failure on anti-PD-1/PD-L1 immune checkpoint inhibitors, with 4/5 patients with MDM2 amplification[283] and 2/3 patients with MDM2 or MDM4 amplification[284] experiencing tumor hyperprogression, amplification of MDM2 or MDM4 was not associated with shorter progression-free survival (PFS) in a retrospective analysis of non-small cell lung cancer (NSCLC) outcomes with immune checkpoint inhibitors (hazard ratio of 1.4, p=0.44)[285]. The latter study reported PFS of >2 months for 5/8 patients with MDM2/MDM4 amplification[285].

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE
MYC

ALTERATION
amplification

POTENTIAL TREATMENT STRATEGIES

— Targeted Therapies —

Limited clinical data indicates that MYC activation may predict sensitivity to the pan-MYC inhibitor OMO-103; a Phase 1 study for patients with solid tumors reported 7 SDs (n=18), including 8% tumor reduction in a patient with pancreas adenocarcinoma[286]. Preclinical data indicate MYC overexpression may predict sensitivity to investigational agents targeting CDK1[287-288], CDK2[289], Aurora kinase A[290-297], Aurora kinase B[298-301], glutaminase[302-305], or BET bromodomain-containing proteins[306-309], as well as agents targeting both HDAC and PI3K[310-312]. Exploratory biomarker analysis in a Phase 2 study reported a PFS benefit associated with a combination of the Aurora A kinase inhibitor alisertib and paclitaxel as second-line therapy for patients with MYC-overexpressed small cell lung cancer, but not for patients without MYC overexpression[313]. A PR was reported for a patient with MYC-amplified invasive ductal breast carcinoma treated with an unspecified Aurora kinase inhibitor and taxol[314].

— Nontargeted Approaches —

MYC amplification has also been suggested to predict response to chemotherapy in patients with breast cancer in some studies[315-316]. Preclinical evidence suggests that colon cancer cells with MYC amplification may be more sensitive to 5-fluorouracil and paclitaxel[317-318].

FREQUENCY & PROGNOSIS

In the TCGA dataset, MYC amplification was observed in 7% and 1.6% of lower grade glioma and glioblastoma (GBM) cases, respectively[40,64], whereas MYC mutation has been reported in <1% of GBM samples[64]. Studies have reported a small number of cases of MYC amplification in anaplastic astrocytoma[319], with one study also reporting elevated c-MYC expression in nucleus and cytoplasm, suggesting a role in tumorigenesis[320]. The effect of MYC on prognosis in glioblastoma is unclear; MYC has been reported to increase sensitivity to radiotherapy and temozolomide in some studies, while other studies have associated MYC expression with increased proliferation, tumor grade, and poor prognosis[321-323]. MYC overexpression in astrocytes resulted in increased cell growth and development of characteristics resembling GBM[324].

FINDING SUMMARY

MYC (c-MYC) encodes a transcription factor that regulates many genes related to cell cycle regulation and cell growth. It is an oncogene and may be activated in as many as 20% of cancers[325]. MYC dysregulation (amplification, overexpression, translocation) has been identified in a number of different cancer types[326]. MYC amplification has been significantly linked with increased mRNA and protein levels and results in the dysregulation of a large number of target genes[325,327-328].

GENE
PIK3R1

ALTERATION
K80fs*26

HGVS VARIANT
NM_181523.3:c.237dup (p.K80Efs*26)

VARIANT CHROMOSOMAL POSITION
chr5:67522738

VARIANT ALLELE FREQUENCY (% VAF)
50.9%

POTENTIAL TREATMENT STRATEGIES

— Targeted Therapies —

On the basis of clinical[329-330] and preclinical[331-332] data, PIK3R1 alteration may predict sensitivity to pan-PI3K or PI3K-alpha-selective inhibitors. In patients with PIK3R1 mutation and no other alterations in the PI3K-AKT-mTOR pathway, 2 CRs have been achieved by patients with endometrial cancer treated with the pan-PI3K inhibitor pilaralisib[329], and 1 PR has been achieved by a patient with breast cancer treated with the PI3K-alpha inhibitor alpelisib in combination with ribociclib and letrozole[333]. Limited clinical and preclinical data suggest that PIK3R1 alterations may also be sensitive to inhibitors of mTOR[332,334-337] or AKT[338-339]. One preclinical study reported that PIK3R1 truncation mutations in the 299–370 range confer sensitivity to MEK inhibitors[340].

FREQUENCY & PROGNOSIS

One study detected PIK3R1 mutation in 10% (24/232) of IDH-wildtype glioblastoma (GBM) samples analyzed[67]. In the TCGA datasets, PIK3R1 mutation is most frequently observed in endometrial carcinoma (33%)[42], glioblastoma (GBM; 11%)[64], uterine carcinosarcoma (11%)(cBioPortal, Jan 2023)[115,197], and lower grade glioma (5%)[106]. PIK3R1 is often inactivated by in-frame insertions or deletions (indels), and the majority of this class of mutation (80%) was observed in endometrial carcinoma[341-343], although PIK3R1 indels have been reported in other cancer types such as GBM, cervical squamous cell carcinoma, and urothelial bladder carcinoma[341]. On the basis of limited clinical data, reduced PIK3R1 expression has been associated with reduced disease-free survival in prostate cancer[344] and metastasis-free survival in breast cancer[345]. PIK3R1 expression is not associated with OS in neuroendocrine tumors[346].

FINDING SUMMARY

PIK3R1 encodes the p85-alpha regulatory subunit of phosphatidylinositol 3-kinase (PI3K)[347]. Loss of PIK3R1 has been shown to result in increased PI3K signaling[348-351], promote tumorigenesis[331,338,348], and promote hyperplasia in the context of PTEN-deficiency[352]. Alterations such as seen here may disrupt PIK3R1 function or expression[332,339-340,342-343,353-362].

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE
KDM5C

ALTERATION
inversion exons 16-19

POTENTIAL TREATMENT STRATEGIES

— Targeted Therapies —

There are no targeted therapies available to address genomic alterations in KDM5C.

FREQUENCY & PROGNOSIS

In a pan-cancer analysis, KDM5C alterations were detected in 2.1% of cases, with the highest frequencies reported in esophagogastric cancer, endometrial cancer, and renal cell carcinoma[363]. Retrospective analysis of 271 nonredundant studies from cBioPortal (n=45,614) showed KDM5C alterations to associate with shorter OS (53 vs. 102 months, HR=1.31)[363]; however, KDM5C alterations have been associated with improved OS for patients treated with immune checkpoint inhibitors across tumor types[363], including non-small cell lung cancer (NSCLC)[364].

FINDING SUMMARY

KDM5C encodes a histone lysine demethylase that acts, along with related histone-modifying enzymes, to control gene expression in response to developmental and environmental cues[365]. In addition to its role as a histone-modifying demethylase, KDM5C has been suggested to play a role in regulation of the SMAD3 signal transduction response to TGF-beta, a role that would be consistent with function as a tumor suppressor[366]. Germline inactivating mutations in KDM5C cause an X-linked intellectual disability syndrome also characterized by short stature and hyperreflexia[367].

GENE
NOTCH1

ALTERATION
E1595fs*13 - subclonal

HGVS VARIANT
NM_017617.3:c.4782_4786del (p.E1595Qfs*13)

VARIANT CHROMOSOMAL POSITION
chr9:139399356-139399361

VARIANT ALLELE FREQUENCY (% VAF)
5.3%

POTENTIAL TREATMENT STRATEGIES

— Targeted Therapies —

NOTCH1 inhibitors and gamma-secretase inhibitors (GSIs) may be potential therapeutic approaches in the case of NOTCH1 activating mutations[368-376]. In a Phase 2 study, the GSI AL101 (BMS-906024) elicited PRs in 15% (6/39) and SDs in 54% (21/39) of patients with metastatic adenoid cystic carcinoma (ACC) harboring NOTCH activating alterations[377]. Additional responses to AL101 have been reported in a patient with gastroesophageal junction adenocarcinoma harboring multiple NOTCH1 mutations, a patient with T-cell acute lymphoblastic leukemia (T-ALL) harboring a NOTCH1 HD domain mutation, and a patient with ACC harboring a single NOTCH1 mutation[378]. A Phase 1 study of the pan-NOTCH inhibitor CB-103 for patients with advanced or recurrent solid tumors reported a preliminary mPFS of 21.7 weeks for patients with ACC, with 2 patients harboring NOTCH1-mutated ACC demonstrating SD > 6 months as best response[379]. While activating mutations may be targeted via gamma-secretase inhibitors or PI3K inhibitors, there are no therapies available to address NOTCH1 inactivation, as seen here.

FREQUENCY & PROGNOSIS

NOTCH1 mutation has been observed in 11% of lower grade glioma cases analyzed[40,106]. NOTCH1 mutation is rare in glioblastoma and has been observed in 0-1.7% of cases, but has been reported in 19-27% of oligodendrogliomas (COSMIC, cBioPortal, Mar 2023)[64,115,196-197,380-381]. NOTCH1 expression has been observed in 60%, 67-68%, 85-93%, and 92-100% of grades 1, 2, 3, and 4 astrocytoma samples, respectively[382-383]. In one study of 18 pilocytic astrocytomas (PA) and 4 low grade astrocytomas, NOTCH1 mRNA over-expression was observed in 95% (21/22) of cases[384]. In addition, the NOTCH1 target, HES1, exhibited moderate to strong protein expression via immunohistochemistry in 66% (40/61) PA samples compared to non-neoplastic brain samples[384]. NOTCH1 expression was significantly increased from Grade 1 to Grade 4 gliomas and was associated with glioma progression[385], as well as with tumor grade and tumor proliferation activity in astrocytoma[382-383].

FINDING SUMMARY

NOTCH1 encodes a member of the NOTCH family of receptors, which are involved in cell fate determination and various developmental processes. Depending on cellular context, NOTCH1 can act as either a tumor suppressor or an oncogene[386-387]. Upon binding of membrane-bound ligands, the NOTCH1 intracellular domain (NICD) is cleaved and forms part of a transcription factor complex that regulates downstream target genes involved in cell fate determination, proliferation, and apoptosis[388-389]. NOTCH1 alterations that disrupt ligand binding[390-392] or remove the transmembrane domain (amino acids 1736-1756), RAM domain (amino acids 1757-1926), ankyrin repeats (amino acids 1927-2122) and/or transactivation domain (amino acids 2123-2374) that are necessary for NOTCH1 function, such as observed here, are predicted to be inactivating[389,393-395]. Several point mutations, including D469G, A465T, C478F, R1594Q, and P1770S, have also been reported to inactivate NOTCH1[386,396-397].

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE

TERT

ALTERATION
promoter -146C>T

HGVS VARIANT
NM_198253.2:c.-146C>T

VARIANT CHROMOSOMAL POSITION
chr5:1295250

VARIANT ALLELE FREQUENCY (% VAF)
44.4%

POTENTIAL TREATMENT STRATEGIES

— Targeted Therapies —

Therapeutic options for targeting tumors with TERT mutations are limited, although a variety of approaches have been investigated, including immunotherapies using TERT as a tumor-associated antigen and antisense oligonucleotide- or peptide-based therapies. TERT peptide vaccines showed limited anticancer efficacy in clinical trials[398]; however, in one preclinical study, the combination of a TERT peptide vaccine and anti-CTLA-4 therapy suppressed tumor growth[399]. A Phase 2 study of the TERT inhibitor imetelstat for patients with advanced non-small cell lung cancer reported no improvement in PFS or OS[400].

FREQUENCY & PROGNOSIS

TERT promoter mutations have been reported in 51-59% of gliomas[401-402], most frequently in glioblastoma (GBM, 54-84%), gliosarcoma (81%), oligodendroglioma (78%), and historically in oligoastrocytomas (25-31%) but less frequently in lower grade astrocytomas (10-18%) and in only 1% of ependymomas[401-405]. In patients with glioblastoma (GBM), the prevalence of TERT promoter mutation is lower in pediatric primary GBM (11%) and adult secondary GBM (28%) compared with adult primary GBM (58-83%)[401,403]. One study detected TERT promoter mutations in 78% (181/232) of IDH-wildtype GBM samples analyzed[67]. TERT promoter mutation has been shown to be significantly associated with increased TERT gene expression in astrocytoma, oligodendroglioma, and GBM[406]. TERT promoter mutations significantly associate with poor prognosis in patients with GBM, although this correlation may be due to the association with primary GBM as opposed to IDH-positive secondary GBM[401,403,406-407]. In the context of IDH-wildtype glioma, TERT mutations are associated with reduced OS (NCCN CNS Cancers Guidelines, v2.2022).

FINDING SUMMARY

Telomerase reverse transcriptase (TERT, or hTERT) is a catalytic subunit of the telomerase complex, which is required to maintain appropriate chromosomal length[408]. Activation of TERT is a hallmark of cancer, being detected in up to 80-90% of malignancies and absent in quiescent cells[409-411]. Mutations within the promoter region of TERT that confer enhanced TERT promoter activity have been reported in two hotspots, located at -124 bp and -146 bp upstream of the transcriptional start site (also termed C228T and C250T, respectively)[412-414], as well as tandem mutations at positions −124/−125 bp and −138/−139 bp[412].

POTENTIAL DIAGNOSTIC IMPLICATIONS

TERT mutations are associated with 1p/19q co-deletion in oligodendrogliomas, and are highly recurrent in IDH/ATRX-wildtype glioblastoma (GBM) (NCCN CNS Cancers Guidelines, v2.2022)[415]. The presence of EGFR gene amplification or TERT promoter mutations are indicative of diffuse astrocytic glioma with molecular features of glioblastoma, WHO grade 4 in IDH1/2-wildtype tumors (NCCN CNS Cancers Guidelines, v2.2022)[75].

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

Cetuximab

Assay findings association

EGFR
amplification

AREAS OF THERAPEUTIC USE

Cetuximab is a monoclonal antibody that targets EGFR. It is FDA approved for the treatment of head and neck squamous cell carcinoma (HNSCC) and KRAS-wild-type, EGFR-expressing metastatic colorectal cancer (CRC). Please see the drug label for full prescribing information.

GENE ASSOCIATION

For patients with metastatic CRC receiving cetuximab or panitumumab as mono- or combination therapy, increased EGFR copy number associated with improved OS (HR=0.62) in a meta-analysis, although increased survival was not seen in populations that received first-line treatment with EGFR antibodies[57].

SUPPORTING DATA

A Phase 2 trial of cetuximab with the anti-VEGF monoclonal antibody bevacizumab for patients with glioblastoma (GBM) did not show improved efficacy compared with bevacizumab alone[416]. However, another Phase 2 study demonstrated that for patients with GBM harboring EGFR amplification but lacking expression of the EGFRvIII variant, treatment with cetuximab resulted in significantly longer PFS and numerical (although not statistically significant) improvement in OS[417]. In addition, a case report for an EGFR-amplified patient with GBM treated with cetuximab using intraarterial cerebral infusion (SIACI) in combination with chemotherapy reported a stable response without recurrence at 6 months[418].

Imatinib

Assay findings association

KIT
amplification

PDGFRA
amplification

AREAS OF THERAPEUTIC USE

Imatinib targets the BCR-ABL fusion protein, PDGFR, and KIT. It is FDA approved for the treatment of KIT-positive gastrointestinal stromal tumors (GIST), Ph+ chronic myeloid leukemia (CML) and acute lymphoblastic leukemia (ALL), myelodysplastic syndrome/ myeloproliferative syndrome (MDS/MPS), aggressive systemic mastocytosis without a D816V KIT mutation, hypereosinophilic syndrome and/or chronic eosinophilic leukemia, and dermatofibrosarcoma protuberans. Please see the drug label for full prescribing information.

GENE ASSOCIATION

On the basis of clinical and preclinical data in KIT-mutated[89-90,130,419] , KIT-amplified[88-91] , or KIT-expressing tumors[93-98,420-421] , KIT activating alterations may confer sensitivity to imatinib. PDGFRA amplification may predict sensitivity to tyrosine kinase inhibitors such as imatinib; a patient with Merkel cell carcinoma expressing PDGFRA achieved a complete response to imatinib[128].

SUPPORTING DATA

In a clinical study where patients with recurrent glioblastoma were given imatinib, 2/24 patients achieved a PR, 10 patients reported SD, and median OS and PFS was observed to be 6.2 and 3 months, respectively[422]. However, other Phase 2 clinical trials of imatinib have reported no anti-tumor activity, with a study of 231 patients with glioblastoma reporting a radiographic response rate of only 3.4%[98,423] . In another Phase 2 study, imatinib plus hydroxyurea was shown to be well tolerated among patients with recurrent/progressive low-grade glioma, but had negligible antitumor activity[424].

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

Nilotinib

Assay findings association

KIT
amplification

AREAS OF THERAPEUTIC USE

Nilotinib targets tyrosine kinases such as ABL (including BCR-ABL), PDGFRs, KIT, CSF1R, DDR1, and DDR2. It is FDA approved to treat newly diagnosed pediatric or adult patients with Philadelphia chromosome-positive (Ph+) chronic myeloid leukemia (CML) in chronic phase, adults with Ph+ CML in chronic or accelerated phase with resistance or intolerance to prior therapy including imatinib, and pediatric patients with Ph+ CML in chronic phase with resistance or intolerance to prior tyrosine-kinase inhibitor therapy. Please see the drug label for full prescribing information.

GENE ASSOCIATION

On the basis of clinical and preclinical data in KIT-mutated[99,425-428] , KIT-amplified[99], or KIT-expressing tumors[429], KIT activating alterations may confer sensitivity to nilotinib.

SUPPORTING DATA

Clinical data on the efficacy of nilotinib for the treatment of CNS tumors are limited (PubMed, Jan 2023). Nilotinib has been primarily investigated as a therapeutic option for the treatment of chronic myeloid leukemia (CML) or gastrointestinal stromal tumors (GIST). In the context of CML, a Phase 3 clinical trial for patients who are Philadelphia chromosome (Ph) positive treated with imatinib or nilotinib (300 or 400 mg) reported PFS rates of 93% and 97-98% and OS rates of 93% and 94-97%, respectively, at 4 years[430]. For Japanese patients with CML who are resistant to imatinib, a Phase 2 trial reported a 49% major medical response rate to treatment with nilotinib at 12 months[431]. A Phase 3 clinical trial of single-agent nilotinib in 240 patients with advanced GIST who failed prior treatment with imatinib or sunitinib reported no significant difference in PFS between nilotinib and the best supportive care but did report increased OS for patients treated with nilotinib[432]. A Phase 2 trial has shown that nilotinib was well tolerated and suggested it may be particularly useful for treating patients with GIST harboring mutations in KIT exon 17[433].

Panitumumab

Assay findings association

EGFR
amplification

AREAS OF THERAPEUTIC USE

Panitumumab is a monoclonal antibody that targets EGFR. It is FDA approved to treat KRAS wild-type and NRAS wild-type metastatic colorectal cancer (CRC) combined with chemotherapy or as monotherapy for patients who have progressed on prior chemotherapy. Please see the drug label for full prescribing information.

GENE ASSOCIATION

For patients with metastatic CRC receiving cetuximab or panitumumab as mono- or combination therapy, increased EGFR copy number associated with improved OS (HR=0.62) in a meta-analysis, although increased survival was not seen in populations that received first-line treatment with EGFR antibodies[57].

SUPPORTING DATA

A Phase 1 trial of EnGeneIC delivery vehicle (EDV) targeting EGFR with panitumumab in combination with doxorubicin for 14 patients with glioblastoma (GBM) reported no responses and 28% (4/14) SDs[434]. Panitumumab has shown efficacy as monotherapy or in combination with chemotherapy for patients with KRAS-wildtype colorectal cancer[435-437] and has been investigated in a variety of other tumor types. For patients with head and neck squamous cell carcinoma (HNSCC), data are conflicting; some trials of panitumumab in various lines and with different chemotherapy combinations have shown modest benefit[438-440] and others have reported no benefit[441-443] . A Phase 3 study of chemotherapy with or without panitumumab for patients with advanced gastroesophageal cancer was terminated for futility[444]. Trials in a variety of tumor types have failed to show significant benefit for patients, including non-small cell lung cancer (NSCLC)[445-446] ; biliary tract cancers, including cholangiocarcinoma[447-448] ; and renal cell carcinoma (RCC)[449].

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

Sorafenib

Assay findings association

KIT
amplification

AREAS OF THERAPEUTIC USE

Sorafenib is a kinase inhibitor that targets the RAF kinases, KIT, FLT3, RET, VEGFRs, and PDGFRs. It is FDA approved for the treatment of unresectable hepatocellular carcinoma, advanced renal cell carcinoma, and recurrent or metastatic differentiated thyroid carcinoma. Please see the drug label for full prescribing information.

GENE ASSOCIATION

On the basis of clinical and preclinical data in KIT-mutated[450-457] or KIT-expressing tumors[100-103] , KIT activating alterations may predict sensitivity to sorafenib.

SUPPORTING DATA

Phase 2 studies of sorafenib plus temozolomide report limited activity in patients with relapsed glioblastoma multiforme (GBM)[458]. A Phase 1/2 trial of temsirolimus in combination with sorafenib in patients with glioblastoma was terminated at the Phase 2 interim analysis after patients failed to meet the primary endpoint of 6 month progression-free survival[459]. A Phase 2 trial of sorafenib and erlotinib in glioblastoma also did not meet its primary endpoint, and erlotinib clearance was increased by the addition of sorafenib[460]. In a Phase 1 trial in patients with high-grade glioma, the combination of sorafenib with radiation therapy (RT) and temozolomide (TMZ) resulted in increased toxicity and did not result in significant improvement in clinical efficacy compared with RT and TMZ alone[461]. In a clinical study of sorafenib in pediatric patients with low-grade astrocytoma, one patient achieved a partial response (PR), one had stable disease (SD), and 9 patients had progressive disease; this study was terminated early due to unexpectedly high disease progression rates[462].

Sunitinib

Assay findings association

KIT
amplification

AREAS OF THERAPEUTIC USE

Sunitinib is a small-molecule tyrosine kinase inhibitor that targets PDGFRs, VEGFRs, KIT, FLT3, CSF-1R, and RET. It is FDA approved for the treatment of advanced or metastatic pancreatic neuroendocrine tumors, gastrointestinal stromal tumors (GISTs) in patients who have progressed on or are intolerant to imatinib, and advanced renal cell carcinoma (RCC) as well as for the adjuvant treatment of patients at high risk of recurrent RCC after nephrectomy. Please see the drug label for full prescribing information.

GENE ASSOCIATION

On the basis of clinical and preclinical data in KIT-mutated[104,463-467] or KIT-expressing tumors[104-105] , KIT activating alterations may predict sensitivity to sunitinib.

SUPPORTING DATA

Phase 2 clinical trials of sunitinib in glioblastoma have reported no significant improvement in clinical outcome[468-469] . A Phase 2 trial that examined sunitinib treatment followed by radiation therapy in patients with glioblastoma reported a median progression-free survival (PFS) of 7.7 weeks, and a median overall survival (OS) of 12.8 weeks; 83.3% (10/12) of patients experienced neurological deterioration prior to radiation therapy[470]. Another Phase 2 study that examined daily sunitinib treatment in patients with glioblastoma reported no objective response in any of the 40 patients, with a median PFS of 2.2 months and a median OS of 9.2 months; five patients in the study had stable disease for more than six months[471].

NOTE Genomic alterations detected may be associated with activity of certain FDA approved drugs, however, the agents listed in this report may have varied evidence in the patient's tumor type.

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

NOTE Clinical trials are ordered by gene and prioritized by: age range inclusion criteria for pediatric patients, proximity to ordering medical facility, later trial phase, and verification of trial information within the last two months. While every effort is made to ensure the accuracy of the information contained below, the information available in the public domain is continually updated and should be investigated by the physician or research staff. This is not a comprehensive list of all available clinical trials. Foundation Medicine displays a subset of trial options and ranks them in this order of descending priority: Qualification for pediatric trial → Geographical proximity → Later trial phase. Clinical trials listed here may have additional enrollment criteria that may require medical screening to determine final eligibility. For additional information about listed clinical trials or to conduct a search for additional trials, please see clinicaltrials.gov. Or, visit https://www.foundationmedicine.com/genomic-testing#support-services.

GENE
ARID1A

ALTERATION
E1924*

RATIONALE
ARID1A loss or inactivation may predict sensitivity to ATR inhibitors.

NCT02264678

PHASE 1/2

Ascending Doses of AZD6738 in Combination With Chemotherapy and/or Novel Anti Cancer Agents

TARGETS
ATR, PARP, PD-L1

LOCATIONS: Seongnam-si (Korea, Republic of), Seoul (Korea, Republic of), Goyang-si (Korea, Republic of), Cambridge (United Kingdom), Withington (United Kingdom), Manchester (United Kingdom), London (United Kingdom), Coventry (United Kingdom), Sutton (United Kingdom), Oxford (United Kingdom)

NCT04802174

PHASE 1/2

Lurbinectedin With Berzosertib, an ATR Kinase Inhibitor in Small Cell Cancers and High-Grade Neuroendocrine Cancers

TARGETS
ATR

LOCATIONS: Maryland

NCT04657068

PHASE 1/2

A Study of ART0380 for the Treatment of Advanced or Metastatic Solid Tumors

TARGETS
ATR

LOCATIONS: London (United Kingdom), Colorado, Oklahoma, Texas, Pennsylvania, Tennessee, Florida

NCT04514497

PHASE 1

Testing the Addition of an Anti-cancer Drug, BAY 1895344, to Usual Chemotherapy for Advanced Stage Solid Tumors, With a Specific Focus on Patients With Small Cell Lung Cancer, Poorly Differentiated Neuroendocrine Cancer, and Pancreatic Cancer

TARGETS
TOP1, ATR

LOCATIONS: California, Arizona, Minnesota, Oklahoma, Missouri, Pennsylvania, Connecticut, New York

NCT03669601

PHASE 1

AZD6738 & Gemcitabine as Combination Therapy

TARGETS
ATR

LOCATIONS: Cambridge (United Kingdom)

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

NCT04616534

PHASE 1

Testing the Addition of an Anti-cancer Drug, BAY 1895344 ATR Inhibitor, to the Chemotherapy Treatment (Gemcitabine) for Advanced Pancreatic and Ovarian Cancer, and Advanced Solid Tumors

TARGETS
ATR

LOCATIONS: Massachusetts, Maryland

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE
CDK4

ALTERATION
amplification

RATIONALE
CDK4 amplification may predict sensitivity to CDK4/6 inhibitors.

NCT04282031

PHASE 1/2

A Study of BPI-1178 in Patients With Advanced Solid Tumor and HR+/HER2- Breast Cancer

TARGETS
CDK6, CDK4, ER, Aromatase

LOCATIONS: Shanghai (China)

NCT03239015

PHASE 2

Efficacy and Safety of Targeted Precision Therapy in Refractory Tumor With Druggable Molecular Event

TARGETS
EGFR, ERBB4, ERBB2, PARP, mTOR, MET, ROS1, RET, VEGFRs, BRAF, CDK4, CDK6

LOCATIONS: Shanghai (China)

NCT04557449

PHASE 1

Study to Test the Safety and Tolerability of PF-07220060 in Participants With Advance Solid Tumors

TARGETS
CDK4, Aromatase, ER

LOCATIONS: Wuhan (China), Zhengzhou (China), Xi'an (China), Beijing (China), Nove Zamky (Slovakia), Bratislava (Slovakia), Praha 2 (Czechia), Edinburgh (United Kingdom), Manchester (United Kingdom), London (United Kingdom)

NCT04391595

PHASE NULL

LY3214996 Plus Abemaciclib in Recurrent Glioblastoma Patients

TARGETS
CDK4, CDK6, ERK1, ERK2

LOCATIONS: Arizona

NCT05262400

PHASE 1/2

A Study to Learn About the Study Medicine (Called PF-07220060 in Combination With PF-07104091) In Participants With Breast Cancer and Solid Tumors

TARGETS
CDK2, Aromatase, CDK4, ER

LOCATIONS: Prague (Czechia), Washington, California, Michigan, Massachusetts, Texas, Merida (Mexico)

NCT03994796

PHASE 2

Genetic Testing in Guiding Treatment for Patients With Brain Metastases

TARGETS
TRKB, ALK, TRKC, ROS1, TRKA, CDK4, CDK6, PI3K, mTOR

LOCATIONS: Washington, Oregon, Idaho, Montana

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

NCT05252416

PHASE 1/2

(VELA) Study of BLU-222 in Advanced Solid Tumors

TARGETS
ER, CDK4, CDK6, CDK2

LOCATIONS: Illinois, Massachusetts, Arkansas, New York, Virginia, Texas, Florida

NCT05159245

PHASE 2

The Finnish National Study to Facilitate Patient Access to Targeted Anti-cancer Drugs

TARGETS
BRAF, VEGFRs, RET, KIT, ERBB2, TRKB, ALK, TRKC, ROS1, TRKA, SMO, PD-L1, MEK, CDK4, CDK6

LOCATIONS: Kuopio (Finland), Helsinki (Finland), Tampere (Finland), Turku (Finland)

NCT03158389

PHASE 1/2

NCT Neuro Master Match - N^2M^2 (NOA-20)

TARGETS
ALK, RET, CDK4, CDK6, mTOR, MDM2, PD-L1, SMO

LOCATIONS: Berlin (Germany), Dresden (Germany), Regensburg (Germany), Bochum (Germany), Frankfurt am Main (Germany), Essen (Germany), Mainz (Germany), Heidelberg (Germany), Cologne (Germany), Mannheim (Germany)

NCT04116541

PHASE 2

A Study Evaluating the Activity of Anti-cancer Treatments Targeting Tumor Molecular Alterations/ Characteristics in Advanced / Metastatic Tumors.

TARGETS
CDK6, CDK4, MDM2, MET, ROS1, RET, VEGFRs

LOCATIONS: Villejuif (France), Nice (France), Lyon (France), Marseille (France), Toulouse (France), Bordeaux (France)

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE
EGFR

ALTERATION
amplification

RATIONALE
EGFR activating mutations, rearrangements, or amplification may predict sensitivity to EGFR-targeted therapies. Strategies to overcome resistance to current agents include next-generation EGFR inhibitors and combination therapies.

NCT04946968

PHASE 2

Phase-2 Dacomitinib Study on Patients With EGFR-Driven Advanced Solid Tumours With Low EGFR-AS1 lncRNA Expr or Other Novel Emerging Biomarkers

TARGETS
ERBB4, EGFR, ERBB2

LOCATIONS: Singapore (Singapore)

NCT04670679

PHASE 1

A Dose Escalation/Expansion Study of ERAS-601 in Patients With Advanced or Metastatic Solid Tumors

TARGETS
SHP2, EGFR

LOCATIONS: Perth (Australia), Melbourne (Australia), Nevada, California, Missouri, Texas, Massachusetts, New York, Pennsylvania, Tennessee

NCT02800486

PHASE 2

Super Selective Intra-arterial Repeated Infusion of Cetuximab (Erbitux) With Reirradiation for Treatment of Relapsed/Refractory GBM, AA, and AOA

TARGETS
EGFR

LOCATIONS: New York

NCT02861898

PHASE 1/2

Super-selective Intra-arterial Repeated Infusion of Cetuximab for the Treatment of Newly Diagnosed Glioblastoma

TARGETS
EGFR

LOCATIONS: New York

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE
KIT

ALTERATION
amplification

RATIONALE
KIT amplification or activating mutations may predict sensitivity to small molecule tyrosine kinase inhibitors. Also, because KIT activation leads to activation of the PI3K-AKT-mTOR pathway, PI3K and mTOR pathway inhibitors may be relevant in a tumor with KIT activation.

NCT04008797

PHASE 1

A Study of E7386 in Combination With Other Anticancer Drug in Participants With Solid Tumor

TARGETS
CBP, Beta-catenin, FGFRs, RET, PDGFRA, VEGFRs, KIT

LOCATIONS: Taipei (Taiwan), Tainan (Taiwan), Kurume (Japan), Matsuyama (Japan), Seongnamsi Bundang (Korea, Republic of), Songpa-gu (Korea, Republic of), Seodaemun (Korea, Republic of), Seoul (Korea, Republic of), Jongno-gu (Korea, Republic of), Osakasayama (Japan)

NCT05024214

PHASE 1/2

Phase Ib/II Trial of Envafolimab Plus Lenvatinib for Subjects With Solid Tumors

TARGETS
PD-L1, FGFRs, RET, PDGFRA, VEGFRs, KIT, FLT3, CSF1R

LOCATIONS: Hangzhou (China), Shanghai (China), Dongguan (China), Guangzhou (China), Zhuhai (China), Benbu (China), Zhengzhou (China), Jinan (China), Dalian (China), Tianjin (China)

NCT05098847

PHASE 2

Cryoablation Combined With Sintilimab Plus Lenvatinib In Previously Treated Unresectable Liver Metastasis From Solid Tumors

TARGETS
FGFRs, RET, PDGFRA, VEGFRs, KIT, PD-1

LOCATIONS: Shanghai (China)

NCT04977453

PHASE 1/2

GI-101 as a Single Agent or in Combination With Pembrolizumab, Lenvatinib or Local Radiotherapy in Advanced Solid Tumors

TARGETS
FGFRs, RET, PDGFRA, VEGFRs, KIT, PD-1, CTLA-4

LOCATIONS: Daejeon (Korea, Republic of), Suwon-si (Korea, Republic of), Seoul (Korea, Republic of), North Carolina

NCT04803318

PHASE 2

Trametinib Combined With Everolimus and Lenvatinib for Recurrent/Refractory Advanced Solid Tumors

TARGETS
mTOR, FGFRs, RET, PDGFRA, VEGFRs, KIT, MEK

LOCATIONS: Guangzhou (China)

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

NCT03564691

PHASE 1

Study of MK-4830 as Monotherapy and in Combination With Pembrolizumab (MK-3475) in Participants With Advanced Solid Tumors (MK-4830-001)

TARGETS
ITL4, FGFRs, RET, PDGFRA, VEGFRs, KIT, PD-1

LOCATIONS: Seoul (Korea, Republic of), Chengdu (China), Brisbane (Australia), Liverpool (Australia), Petah Tikva (Israel), Ramat Gan (Israel), Tel Aviv (Israel), Haifa (Israel), Warszawa (Poland), Gdansk (Poland)

NCT05740215

PHASE 1/2

Efficacy and Safety Study of F520 Combined With Lenvatinib in the Treatment of Patients With Advanced Solid Tumors

TARGETS
PD-1, FGFRs, RET, PDGFRA, VEGFRs, KIT

LOCATIONS: Chongqing (China)

NCT05064280

PHASE 2

Phase II Study of Pembrolizumab in Combination With Lenvatinib in Patients With TNBC, NSCLC, and Other Tumor Types and Brain Metastases

TARGETS
PD-1, KIT, VEGFRs, FGFRs, PDGFRA, RET

LOCATIONS: Texas

NCT04817956

PHASE 2

Improving Public Cancer Care by Implementing Precision Medicine in Norway

TARGETS
PD-L1, VEGFA, ERBB2, ALK, RET, PARP, SMO, TRKB, TRKC, ROS1, TRKA, MEK, BRAF, PI3K-alpha, FGFR1, FGFR2, FGFR3, MET, KIT, ABL

LOCATIONS: Tromsø (Norway), Bodø (Norway), Hamar (Norway), Oslo (Norway), Fredrikstad (Norway), Drammen (Norway), Trondheim (Norway), Skien (Norway), Førde (Norway), Bergen (Norway)

NCT04729348

PHASE 2

Pembrolizumab And Lenvatinib In Leptomeningeal Metastases

TARGETS
PD-1, KIT, VEGFRs, FGFRs, PDGFRA, RET

LOCATIONS: Massachusetts

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE
MDM2

ALTERATION
amplification

RATIONALE
Inhibitors of the MDM2-p53 interaction are being tested in clinical trials. Overexpression or amplification of MDM2 may increase sensitivity to these agents, but more data are required.

NCT04589845

PHASE 2

Tumor-Agnostic Precision Immuno-Oncology and Somatic Targeting Rational for You (TAPISTRY) Platform Study

TARGETS
TRKB, ALK, TRKC, ROS1, TRKA, RET, PD-L1, AKTs, ERBB2, MDM2, PI3K-alpha, RAFs, NRAS

LOCATIONS: Taipei City (Taiwan), Taoyuan County (Taiwan), Tainan (Taiwan), Shanghai City (China), Shanghai (China), Shatin (Hong Kong), Hong Kong (Hong Kong), Seoul (Korea, Republic of), Seongnam-si (Korea, Republic of), Xi'an (China)

NCT04785196

PHASE 1/2

APG-115 in Combination With PD-1 Inhibitor in Patients With Advanced Liposarcoma or Advanced Solid Tumors

TARGETS
PD-1, MDM2

LOCATIONS: Shanghai (China), Guangzhou (China)

NCT03449381

PHASE 1

This Study Aims to Find the Best Dose of BI 907828 in Patients With Different Types of Advanced Cancer (Solid Tumors)

TARGETS
MDM2

LOCATIONS: Tokyo, Chuo-ku (Japan), Stockholm (Sweden), Warsaw (Poland), Poznan (Poland), Berlin (Germany), Göttingen (Germany), Köln (Germany), Tübingen (Germany), Leuven (Belgium), Bruxelles (Belgium)

NCT03611868

PHASE 1/2

A Study of APG-115 in Combination With Pembrolizumab in Patients With Metastatic Melanomas or Advanced Solid Tumors

TARGETS
MDM2, PD-1

LOCATIONS: Brisbane (Australia), South Brisbane (Australia), Bedford Park (Australia), Heidelberg (Australia), California, Arizona, Missouri, Arkansas, Ohio, Pennsylvania

NCT03158389

PHASE 1/2

NCT Neuro Master Match - N^2M^2 (NOA-20)

TARGETS
ALK, RET, CDK4, CDK6, mTOR, MDM2, PD-L1, SMO

LOCATIONS: Berlin (Germany), Dresden (Germany), Regensburg (Germany), Bochum (Germany), Frankfurt am Main (Germany), Essen (Germany), Mainz (Germany), Heidelberg (Germany), Cologne (Germany), Mannheim (Germany)

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE
MYC

ALTERATION
amplification

RATIONALE
MYC overexpression may predict sensitivity to inhibition of CDKs, especially CDK1 and CDK2, of Aurora kinases, including Aurora kinase A and B, and of BET domain proteins, which are reported to downregulate MYC expression and MYC-dependent transcriptional programs.

NCT05253053

PHASE 1/2

Study to Evaluate the Efficacy and Safety of TT-00420 as Monotherapy and Combination Therapy in Patients With Advanced Solid Tumors

TARGETS
Aurora kinase A, Aurora kinase B, PD-L1

LOCATIONS: Jinan (China), Beijing (China)

NCT04983810

PHASE 1/2

A Study to Investigate Fadraciclib (CYC065), in Subjects With Advanced Solid Tumors and Lymphoma

TARGETS
CDK2, CDK9

LOCATIONS: Seoul (Korea, Republic of), Barcelona (Spain), California, Texas

NCT05252390

PHASE 1/2

NUV-868 as Monotherapy and in Combination With Olaparib or Enzalutamide in Adult Patients With Advanced Solid Tumors

TARGETS
BRD4, PARP, AR

LOCATIONS: Montana, California, Colorado, Arizona, Michigan, Texas

NCT05327010

PHASE 2

Testing the Combination of the Anti-cancer Drugs ZEN003694 (ZEN-3694) and Talazoparib in Patients With Advanced Solid Tumors, The ComBET Trial

TARGETS
PARP, BRD4, BRDT, BRD2, BRD3

LOCATIONS: Colorado, Illinois, Texas, North Carolina, Georgia

NCT04742959

PHASE 1/2

Crossover Relative Bioavailability and Dose Escalation Study of TT-00420 Tablet in Patients With Advanced Solid Tumors

TARGETS
Aurora kinase A, Aurora kinase B

LOCATIONS: California, Illinois, Ohio, Texas, New Jersey

NCT05053971

PHASE 1/2

Testing A New Anti-cancer Drug Combination, Entinostat and ZEN003694, for Advanced and Refractory Solid Tumors and Lymphomas

TARGETS
BRD3, BRD4, BRD2, BRDT, HDAC

LOCATIONS: Oklahoma, Connecticut, Florida

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

NCT04840589

PHASE 1

Testing the Combination of ZEN003694 and Nivolumab With or Without Ipilimumab in Solid Tumors

TARGETS
PD-1, CTLA-4, BRD4, BRDT, BRD2, BRD3

LOCATIONS: Ohio, Pennsylvania, New York, Maryland

NCT04555837

PHASE 1/2

Alisertib and Pembrolizumab for the Treatment of Patients With Rb-deficient Head and Neck Squamous Cell Cancer

TARGETS
Aurora kinase A, PD-1

LOCATIONS: Texas

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE
PDGFRA

ALTERATION
amplification

RATIONALE
PDGFRA amplification may predict sensitivity to imatinib and to anti-PDGFRA antibodies.

NCT04817956

PHASE 2

Improving Public Cancer Care by Implementing Precision Medicine in Norway

TARGETS
PD-L1, VEGFA, ERBB2, ALK, RET, PARP, SMO, TRKB, TRKC, ROS1, TRKA, MEK, BRAF, PI3K-alpha, FGFR1, FGFR2, FGFR3, MET, KIT, ABL

LOCATIONS: Tromsø (Norway), Bodø (Norway), Hamar (Norway), Oslo (Norway), Fredrikstad (Norway), Drammen (Norway), Trondheim (Norway), Skien (Norway), Førde (Norway), Bergen (Norway)

NCT01738139

PHASE 1

Ipilimumab and Imatinib Mesylate in Advanced Cancer

TARGETS
KIT, ABL, CTLA-4

LOCATIONS: Texas

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

GENE
PIK3R1

ALTERATION
K80fs*26

RATIONALE
On the basis of clinical and strong preclinical data, PIK3R1 loss or inactivation may indicate sensitivity to pan-PI3K or PI3K-alpha-selective inhibitors.

NCT04526470

PHASE 1/2

Alpelisib and Paclitaxel in PIK3CA-altered Gastric Cancer

TARGETS
PI3K-alpha

LOCATIONS: Seongnam-si (Korea, Republic of), Seoul (Korea, Republic of)

NCT04817956

PHASE 2

Improving Public Cancer Care by Implementing Precision Medicine in Norway

TARGETS
PD-L1, VEGFA, ERBB2, ALK, RET, PARP, SMO, TRKB, TRKC, ROS1, TRKA, MEK, BRAF, PI3K-alpha, FGFR1, FGFR2, FGFR3, MET, KIT, ABL

LOCATIONS: Tromsø (Norway), Bodø (Norway), Hamar (Norway), Oslo (Norway), Fredrikstad (Norway), Drammen (Norway), Trondheim (Norway), Skien (Norway), Førde (Norway), Bergen (Norway)

NCT04975958

PHASE 1

Double/Triple Combinations of AN2025, AN0025 and Atezolizumab in Advanced Solid Tumors

TARGETS
PI3K, PD-L1, EP4

LOCATIONS: Colorado, Oklahoma, New York, New Jersey, Florida

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

NOTE One or more variants of unknown significance (VUS) were detected in this patient's tumor. These variants may not have been adequately characterized in the scientific literature at the time this report was issued, and/or the genomic context of these alterations makes their significance unclear. We choose to include them here in the event that they become clinically meaningful in the future.

BCORL1
NM_021946.4: c.4091G>A
(p.R1364H)
chrX:129162622

EGFR
NM_005228.3: c.970C>T
(p.R324C)
chr7:55223603 and
rearrangement

KDM6A
NM_021140.2: c.3954A>G
(p.I1318M)
chrX:44966730

KDR
NM_002253.2: c.2498G>A
(p.R833Q)
chr4:55964315

KIT
NM_000222.2: c.1202C>G
(p.A401G)
chr4:55575676 and
NM_000222.2: c.1132C>T
(p.H378Y)
chr4:55575606

MAP3K1
amplification

PBRM1
NM_018313.4: c.2504G>A
(p.R835H)
chr3:52643392

PDGFRB
NM_002609.3: c.1108C>T
(p.R370C)
chr5:149512332

PIK3R1
rearrangement

RAC1
NM_006908.4: c.308_316del
(p.H103_C105del)
chr7:6441516-6441525 and
rearrangement

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

FoundationOne CDx is designed to include genes known to be somatically altered in human solid tumors that are validated targets for therapy, either approved or in clinical trials, and/or that are unambiguous drivers of oncogenesis based on current knowledge. The current assay interrogates 324 genes as well as introns of 36 genes involved in rearrangements. The assay will be updated periodically to reflect new knowledge about cancer biology.

DNA GENE LIST: ENTIRE CODING SEQUENCE FOR THE DETECTION OF BASE SUBSTITUTIONS, INSERTION/DELETIONS, AND COPY NUMBER ALTERATIONS

ABL1	*ACVR1B*	*AKT1*	*AKT2*	*AKT3*	*ALK*	*ALOX12B*	*AMER1 (FAM123B or WTX)*	
APC	*AR*	*ARAF*	*ARFRP1*	*ARID1A*	*ASXL1*	*ATM*	*ATR*	*ATRX*
AURKA	*AURKB*	*AXIN1*	*AXL*	*BAP1*	*BARD1*	*BCL2*	*BCL2L1*	*BCL2L2*
BCL6	*BCOR*	*BCORL1*	*BRAF*	*BRCA1*	*BRCA2*	*BRD4*	*BRIP1*	*BTG1*
BTG2	*BTK*	*CALR*	*CARD11*	*CASP8*	*CBFB*	*CBL*	*CCND1*	*CCND2*
CCND3	*CCNE1*	*CD22*	*CD274 (PD-L1)*	*CD70*	*CD79A*	*CD79B*	*CDC73*	*CDH1*
CDK12	*CDK4*	*CDK6*	*CDK8*	*CDKN1A*	*CDKN1B*	*CDKN2A*	*CDKN2B*	*CDKN2C*
CEBPA	*CHEK1*	*CHEK2*	*CIC*	*CREBBP*	*CRKL*	*CSF1R*	*CSF3R*	*CTCF*
CTNNA1	*CTNNB1*	*CUL3*	*CUL4A*	*CXCR4*	*CYP17A1*	*DAXX*	*DDR1*	*DDR2*
DIS3	*DNMT3A*	*DOT1L*	*EED*	*EGFR*	*EMSY (C11orf30)*	*EP300*	*EPHA3*	*EPHB1*
EPHB4	*ERBB2*	*ERBB3*	*ERBB4*	*ERCC4*	*ERG*	*ERRFI1*	*ESR1*	*EZH2*
FANCA	*FANCC*	*FANCG*	*FANCL*	*FAS*	*FBXW7*	*FGF10*	*FGF12*	*FGF14*
FGF19	*FGF23*	*FGF3*	*FGF4*	*FGF6*	*FGFR1*	*FGFR2*	*FGFR3*	*FGFR4*
FH	*FLCN*	*FLT1*	*FLT3*	*FOXL2*	*FUBP1*	*GABRA6*	*GATA3*	*GATA4*
GATA6	*GID4 (C17orf39)*	*GNA11*	*GNA13*	*GNAQ*	*GNAS*	*GRM3*	*GSK3B*	*H3-3A (H3F3A)*
HDAC1	*HGF*	*HNF1A*	*HRAS*	*HSD3B1*	*ID3*	*IDH1*	*IDH2*	*IGF1R*
IKBKE	*IKZF1*	*INPP4B*	*IRF2*	*IRF4*	*IRS2*	*JAK1*	*JAK2*	*JAK3*
JUN	*KDM5A*	*KDM5C*	*KDM6A*	*KDR*	*KEAP1*	*KEL*	*KIT*	*KLHL6*
KMT2A (MLL)	*KMT2D (MLL2)*	*KRAS*	*LTK*	*LYN*	*MAF*	*MAP2K1 (MEK1)*	*MAP2K2 (MEK2)*	*MAP2K4*
MAP3K1	*MAP3K13*	*MAPK1*	*MCL1*	*MDM2*	*MDM4*	*MED12*	*MEF2B*	*MEN1*
MERTK	*MET*	*MITF*	*MKNK1*	*MLH1*	*MPL*	*MRE11 (MRE11A)*	*MSH2*	*MSH3*
MSH6	*MST1R*	*MTAP*	*MTOR*	*MUTYH*	*MYC*	*MYCL (MYCL1)*	*MYCN*	*MYD88*
NBN	*NF1*	*NF2*	*NFE2L2*	*NFKBIA*	*NKX2-1*	*NOTCH1*	*NOTCH2*	*NOTCH3*
NPM1	*NRAS*	*NSD2 (WHSC1 or MMSET)*		*NSD3 (WHSC1L1)*	*NT5C2*	*NTRK1*	*NTRK2*	*NTRK3*
P2RY8	*PALB2*	*PARP1*	*PARP2*	*PARP3*	*PAX5*	*PBRM1*	*PDCD1 (PD-1)*	*PDCD1LG2 (PD-L2)*
PDGFRA	*PDGFRB*	*PDK1*	*PIK3C2B*	*PIK3C2G*	*PIK3CA*	*PIK3CB*	*PIK3R1*	*PIM1*
PMS2	*POLD1*	*POLE*	*PPARG*	*PPP2R1A*	*PPP2R2A*	*PRDM1*	*PRKAR1A*	*PRKCI*
PRKN (PARK2)	*PTCH1*	*PTEN*	*PTPN11*	*PTPRO*	*QKI*	*RAC1*	*RAD21*	*RAD51*
RAD51B	*RAD51C*	*RAD51D*	*RAD52*	*RAD54L*	*RAF1*	*RARA*	*RB1*	*RBM10*
REL	*RET*	*RICTOR*	*RNF43*	*ROS1*	*RPTOR*	*SDHA*	*SDHB*	*SDHC*
SDHD	*SETD2*	*SF3B1*	*SGK1*	*SMAD2*	*SMAD4*	*SMARCA4*	*SMARCB1*	*SMO*
SNCAIP	*SOCS1*	*SOX2*	*SOX9*	*SPEN*	*SPOP*	*SRC*	*STAG2*	*STAT3*
STK11	*SUFU*	*SYK*	*TBX3*	*TEK*	*TENT5C (FAM46C)*		*TET2*	*TGFBR2*
TIPARP	*TNFAIP3*	*TNFRSF14*	*TP53*	*TSC1*	*TSC2*	*TYRO3*	*U2AF1*	*VEGFA*
VHL	*WT1*	*XPO1*	*XRCC2*	*ZNF217*	*ZNF703*			

DNA GENE LIST: FOR THE DETECTION OF SELECT REARRANGEMENTS

ALK	*BCL2*	*BCR*	*BRAF*	*BRCA1*	*BRCA2*	*CD74*	*EGFR*	*ETV4*
ETV5	*ETV6*	*EWSR1*	*EZR*	*FGFR1*	*FGFR2*	*FGFR3*	*KIT*	*KMT2A (MLL)*
MSH2	*MYB*	*MYC*	*NOTCH2*	*NTRK1*	*NTRK2*	*NUTM1*	*PDGFRA*	*RAF1*
RARA	*RET*	*ROS1*	*RSPO2*	*SDC4*	*SLC34A2*	*TERC**	*TERT***	*TMPRSS2*

*TERC is an NCRNA

**Promoter region of TERT is interrogated

ADDITIONAL ASSAYS: FOR THE DETECTION OF SELECT CANCER BIOMARKERS

Homologous Recombination status
Loss of Heterozygosity (LOH) score
Microsatellite (MS) status
Tumor Mutational Burden (TMB)

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

FoundationOne CDx fulfills the requirements of the European Directive 98/79 EC for in vitro diagnostic medical devices and is registered as a CE-IVD product by Foundation Medicine's EU Authorized Representative, Qarad b.v.b.a, Cipalstraat 3, 2440 Geel, Belgium. CE

ABOUT FOUNDATIONONE CDX

FoundationOne CDx was developed and its performance characteristics determined by Foundation Medicine, Inc. (Foundation Medicine). FoundationOne CDx may be used for clinical purposes and should not be regarded as purely investigational or for research only. Foundation Medicine's clinical reference laboratories are qualified to perform high-complexity clinical testing.

Please refer to technical information for performance specification details: www.rochefoundationmedicine.com/f1cdxtech.

INTENDED USE

FoundationOne®CDx (F1CDx) is a next generation sequencing based in vitro diagnostic device for detection of substitutions, insertion and deletion alterations (indels), and copy number alterations (CNAs) in 324 genes and select gene rearrangements, as well as genomic signatures including microsatellite instability (MSI), tumor mutational burden (TMB), and for selected forms of ovarian cancer, loss of heterozygosity (LOH) score, using DNA isolated from formalin-fixed, paraffin-embedded (FFPE) tumor tissue specimens. The test is intended as a companion diagnostic to identify patients who may benefit from treatment with therapies in accordance with approved therapeutic product labeling. Additionally, F1CDx is intended to provide tumor mutation profiling to be used by qualified health care professionals in accordance with professional guidelines in oncology for patients with solid malignant neoplasms.

TEST PRINCIPLE

FoundationOne CDx will be performed exclusively as a laboratory service using DNA extracted from formalin-fixed, paraffin-embedded (FFPE) tumor samples. The proposed assay will employ a single DNA extraction method from routine FFPE biopsy or surgical resection specimens, 50-1000 ng of which will undergo whole-genome shotgun library construction and hybridization-based capture of all coding exons from 309 cancer-related genes, one promoter region, one non-coding (ncRNA), and select intronic regions from 34 commonly rearranged genes, 21 of which also include the coding exons. The assay therefore includes detection of alterations in a total of 324 genes.

Using an Illumina® HiSeq platform, hybrid capture–selected libraries will be sequenced to high uniform depth (targeting >500X median coverage with >99% of exons at coverage >100X). Sequence data will be processed using a customized analysis pipeline designed to accurately detect all classes of genomic alterations, including base substitutions, indels, focal copy number amplifications, homozygous gene deletions, and selected genomic rearrangements (e.g.,gene fusions). Additionally, genomic signatures including loss of heterozygosity (LOH), microsatellite instability (MSI) and tumor mutational burden (TMB) will be reported.

THE REPORT

Incorporates analyses of peer-reviewed studies and other publicly available information identified by Foundation Medicine; these analyses and information may include associations between a molecular alteration (or lack of alteration) and one or more drugs with potential clinical benefit (or potential lack of clinical benefit), including drug candidates that are being studied in clinical research. The F1CDx report may be used as an aid to inform molecular eligibility for clinical trials. Note: A finding of biomarker alteration does not necessarily indicate pharmacologic effectiveness (or lack thereof) of any drug or treatment regimen; a finding of no biomarker alteration does not necessarily indicate lack of pharmacologic effectiveness (or effectiveness) of any drug or treatment regimen.

Diagnostic Significance

FoundationOne CDx identifies alterations to select cancer-associated genes or portions of genes (biomarkers). In some cases, the Report also highlights selected negative test results regarding biomarkers of clinical significance.

Qualified Alteration Calls (Equivocal and Subclonal)

An alteration denoted as "amplification – equivocal" implies that the FoundationOne CDx assay data provide some, but not unambiguous, evidence that the copy number of a gene exceeds the threshold for identifying copy number amplification. The threshold used in FoundationOne CDx for identifying a copy number amplification is four (4) for *ERBB2* and six (6) for all other genes. Conversely, an alteration denoted as "loss – equivocal" implies that the FoundationOne CDx assay data provide some, but not unambiguous, evidence for homozygous deletion of the gene in question. An alteration denoted as "subclonal" is one that the FoundationOne CDx analytical methodology has identified as being present in <10% of the assayed tumor DNA.

Ranking of Therapies and Clinical Trials

Ranking of Therapies in Summary Table
Therapies are ranked based on the following criteria: Therapies with clinical benefit (ranked alphabetically within each evidence category), followed by therapies associated with resistance (when applicable).

Ranking of Clinical Trials
Pediatric trial qualification → Geographical proximity → Later trial phase.

NATIONAL COMPREHENSIVE CANCER NETWORK® (NCCN®) CATEGORIZATION

Biomarker and genomic findings detected may be associated with certain entries within the NCCN Drugs & Biologics Compendium® (NCCN Compendium®) (www.nccn.org). The NCCN Categories of Evidence and Consensus indicated reflect the highest possible category for a given therapy in association with each biomarker or genomic finding. Please note, however, that the accuracy and applicability of these NCCN categories within a report may be impacted by the patient's clinical history, additional biomarker information, age, and/or co-occurring alterations. For additional information on the NCCN categories, please refer to the NCCN Compendium®. Referenced with permission from the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®). © National Comprehensive Cancer Network, Inc. 2023. All rights reserved. To view the most recent and complete version of the guidelines, go online to NCCN.org. NCCN makes no warranties of any kind whatsoever regarding their content, use or application and disclaims any responsibility for their application or use in any way.

Limitations

1. In the fraction-based MSI algorithm, a tumor specimen will be categorized as MSI-H, MSS, or MS-Equivocal according to the fraction of microsatellite loci determined to be altered or unstable (i.e., the fraction unstable loci score). In the F1CDx assay, MSI is evaluated based on a genome-wide analysis across >2000 microsatellite loci. For a given microsatellite locus, non-somatic alleles are discarded, and the microsatellite is categorized as unstable if remaining alleles differ from the reference genome. The final fraction unstable loci score is calculated as the number of unstable microsatellite loci divided by the number of evaluable microsatellite loci. The MSI-H and MSS cut-off thresholds were determined by

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

analytical concordance to a PCR comparator assay using a pan-tumor FFPE tissue sample set. Patients with results categorized as "MS-Stable" with median exon coverage <300X, "MS-Equivocal," or "Cannot Be Determined" should receive confirmatory testing using a validated orthogonal (alternative) method.

2. TMB by F1CDx is determined by counting all synonymous and non-synonymous variants present at 5% allele frequency or greater (after filtering) and the total number is reported as mutations per megabase (mut/Mb) unit. Observed TMB is dependent on characteristics of the specific tumor focus tested for a patient (e.g., primary vs. metastatic, tumor content) and the testing platform used for the detection; therefore, observed TMB results may vary between different specimens for the same patient and between detection methodologies employed on the same sample. The TMB calculation may differ from TMB calculations used by other assays depending on variables such as the amount of genome interrogated, percentage of tumor, assay limit of detection (LoD), filtering of alterations included in the score, and the read depth and other bioinformatic test specifications. Refer to the SSED for a detailed description of these variables in FMI's TMB calculation https://www.accessdata.fda.gov/cdrh_docs/pdf17/P170019B.pdf. The clinical validity of TMB defined by this panel has been established for TMB as a qualitative output for a cut-off of 10 mutations per megabase but has not been established for TMB as a quantitative score.
3. Homologous Recombination status may be reported for epithelial ovarian, peritoneal, or Fallopian tube carcinomas (Coleman et al., 2017; 28916367). Samples with deleterious *BRCA1/2* alteration and/or Loss of Heterozygosity (LOH) score ≥ 16% will be reported as "HRD Positive" and samples with absence of these findings will be reported as "HRD Not Detected," agnostic of potential secondary *BRCA1/2* reversion alterations. Certain potentially deleterious missense or small in-frame deletions in *BRCA1/2* may not be classified as deleterious and, in the absence of an elevated LOH profile, samples with such mutations may be classified as "HRD Not Detected." A result of "HRD Not Detected" does not rule out the presence of a *BRCA1/2* alteration or an elevated LOH profile outside the assay performance characteristic limitations.
4. The LOH score is determined by analyzing SNPs spaced at 1Mb intervals across the genome on the FoundationOne CDx test and extrapolating an LOH profile, excluding arm- and chromosome-wide LOH segments. Detection of LOH has been verified only for ovarian cancer patients, and the LOH score result may be reported for epithelial ovarian, peritoneal, or Fallopian tube carcinomas. The LOH score will be reported as "Cannot Be Determined" if the sample is not of sufficient quality to confidently determine LOH. Performance of the LOH classification has not been established for samples below 35% tumor content. There may be potential interference of ethanol with LOH detection. The interfering effects of xylene, hemoglobin, and triglycerides on the LOH score have not been demonstrated.
5. Alterations reported may include somatic (not inherited) or germline (inherited) alterations; however, the test does not distinguish between germline and somatic alterations. The test does not provide information about susceptibility.
6. Biopsy may pose a risk to the patient when archival tissue is not available for use with the assay. The patient's physician should determine whether the patient is a candidate for biopsy.
7. Reflex testing to an alternative FDA approved companion diagnostic should be performed for patients who have an *ERBB2* amplification result detected with copy number equal to 4 (baseline ploidy of tumor +2) for confirmatory testing. While this result is considered negative by FoundationOne®CDx (F1CDx), in a clinical concordance study with an FDA approved FISH test, 70% (7 out of 10 samples) were positive, and 30% (3 out of 10 samples) were negative by the FISH test with an average ratio of 2.3. The frequency of *ERBB2* copy number 4 in breast cancer is estimated to be approximately 2%. Multiple references listed in https://www.mycancergenome.org/content/disease/breast-cancer/ERBB2/238/ report the frequency of HER2 overexpression as 20% in breast cancer. Based on the F1CDx HER2 CDx concordance study, approximately 10% of HER2 amplified samples had copy number 4. Thus, total frequency is conservatively estimated to be approximately 2%.

REPORT HIGHLIGHTS

The Report Highlights includes select genomic and therapeutic information with potential impact on patient care and treatment that is specific to the genomics and tumor type of the sample analyzed. This section may highlight information including targeted therapies with potential sensitivity or resistance; evidence-matched clinical trials; and variants with potential diagnostic, prognostic, nontargeted treatment, germline, or clonal hematopoiesis implications. Information included in the Report Highlights is expected to evolve with advances in scientific and clinical research. Findings included in the Report Highlights should be considered in the context of all other information in this report and other relevant patient information. Decisions on patient care and treatment are the responsibility of the treating physician.

VARIANT ALLELE FREQUENCY

Variant Allele Frequency (VAF) represents the fraction of sequencing reads in which the variant is observed. This attribute is not taken into account for therapy inclusion, clinical trial matching, or interpretive content. Caution is recommended in interpreting VAF to indicate the potential germline or somatic origin of an alteration, recognizing that tumor fraction and tumor ploidy of samples may vary.

Precision of VAF for base substitutions and indels

BASE SUBSTITUTIONS	%CV*
Repeatability	5.11 - 10.40
Reproducibility	5.95 - 12.31
INDELS	**%CV***
Repeatability	6.29 - 10.00
Reproducibility	7.33 - 11.71

**Interquartile Range = 1st Quartile to 3rd Quartile*

VARIANTS TO CONSIDER FOR FOLLOW-UP GERMLINE TESTING

The variants indicated for consideration of follow-up germline testing are 1) limited to reportable short variants with a protein effect listed in the ClinVar genomic database (Landrum et al., 2018; 29165669) as Pathogenic, Pathogenic/Likely Pathogenic, or Likely Pathogenic (by an expert panel or multiple submitters), 2) associated with hereditary cancer-predisposing disorder(s), 3) detected at an allele frequency of >10%, and 4) in select genes reported by the ESMO Precision Medicine Working Group (Mandelker et al., 2019; 31050713) to have a greater than 10% probability of germline origin if identified during tumor sequencing. The selected genes are *ATM, BAP1, BRCA1, BRCA2, BRIP1, CHEK2, FH, FLCN, MLH1, MSH2, MSH6, MUTYH, PALB2, PMS2, POLE, RAD51C, RAD51D, RET, SDHA, SDHB, SDHC, SDHD, TSC2,* and *VHL*, and are not inclusive of all cancer susceptibility genes. The content in this report should not substitute for genetic counseling or follow-up germline testing, which is needed to distinguish whether a finding in this patient's

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

tumor sequencing is germline or somatic. Interpretation should be based on clinical context.

VARIANTS THAT MAY REPRESENT CLONAL HEMATOPOIESIS

Variants that may represent clonal hematopoiesis (CH) are limited to select reportable short variants in defined genes identified in solid tumors only. Variant selection was determined based on gene tumor-suppressor or oncogene status, known role in solid tumors versus hematological malignancies, and literature prevalence. The defined genes are *ASXL1, CBL, DNMT3A, IDH2, JAK2, KMT2D (MLL2), MPL, MYD88, SF3B1, TET2,* and *U2AF1* and are not inclusive of all CH genes. The content in this report should not substitute for dedicated hematological workup. Comprehensive genomic profiling of solid tumors detects nontumor alterations that are due to CH. Patient-matched peripheral blood mononuclear cell sequencing is required to conclusively determine if this alteration is present in tumor or is secondary to CH. Interpretation should be based on clinical context.

LEVEL OF EVIDENCE NOT PROVIDED

Drugs with potential clinical benefit (or potential lack of clinical benefit) are not evaluated for source or level of published evidence.

NO GUARANTEE OF CLINICAL BENEFIT

This Report makes no promises or guarantees that a particular drug will be effective in the treatment of disease in any patient. This Report also makes no promises or guarantees that a drug with potential lack of clinical benefit will in fact provide no clinical benefit.

NO GUARANTEE OF REIMBURSEMENT

Foundation Medicine makes no promises or guarantees that a healthcare provider, insurer or other third party payor, whether private or governmental, will reimburse a patient for the cost of FoundationOne CDx.

TREATMENT DECISIONS ARE RESPONSIBILITY OF PHYSICIAN

Drugs referenced in this Report may not be suitable for a particular patient. The selection of any, all or none of the drugs associated with potential clinical benefit (or potential lack of clinical benefit) resides entirely within the discretion of the treating physician. Indeed, the information in this Report must be considered in conjunction with all other relevant information regarding a particular patient, before the patient's treating physician recommends a course of treatment. Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all applicable information concerning the patient's condition, such as patient and family history, physical examinations, information from other diagnostic tests, and patient preferences, in accordance with the standard of care in a given community. A treating physician's decisions should not be based on a single test, such as this Test, or the information contained in this Report. Certain sample or variant characteristics may result in reduced sensitivity. FoundationOne CDx is performed using DNA derived from tumor, and as such germline events may not be reported.

SELECT ABBREVIATIONS

ABBREVIATION	DEFINITION
CR	Complete response
DCR	Disease control rate
DNMT	DNA methyltransferase
HR	Hazard ratio
ITD	Internal tandem duplication
MMR	Mismatch repair
muts/Mb	Mutations per megabase
NOS	Not otherwise specified
ORR	Objective response rate
OS	Overall survival
PD	Progressive disease
PFS	Progression-free survival
PR	Partial response
SD	Stable disease
TKI	Tyrosine kinase inhibitor

REFERENCE SEQUENCE INFORMATION

Sequence data is mapped to the human genome, Genome Reference Consortium Human Build 37 (GRCh37), also known as hg19.

MR Suite Version (RG) 7.9.0

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of the transformed content, and FMI does not provide any warranty to validate the suitability of use.

The median exon coverage for this sample is 751x

© 2023 Foundation Medicine, Inc. All rights reserved.

1. Gatalica Z, et al. Cancer Epidemiol. Biomarkers Prev. (2014) pmid: 25392179
2. Kroemer G, et al. Oncoimmunology (2015) pmid: 26140250
3. Lal N, et al. Oncoimmunology (2015) pmid: 25949894
4. Le DT, et al. N. Engl. J. Med. (2015) pmid: 26028255
5. Ayers et al., 2016; ASCO-SITC Abstract P60
6. Martinez R, et al. Oncology (2004) pmid: 15331927
7. Martinez R, et al. J. Cancer Res. Clin. Oncol. (2005) pmid: 15672285
8. Martinez R, et al. Cancer Genet. Cytogenet. (2007) pmid: 17498554
9. Szybka M, et al. Clin. Neuropathol. () pmid: 12908754
10. Kocarnik JM, et al. Gastroenterol Rep (Oxf) (2015) pmid: 26337942
11. You JF, et al. Br. J. Cancer (2010) pmid: 21081928
12. Bairwa NK, et al. Methods Mol. Biol. (2014) pmid: 24623249
13. Boland CR, et al. Cancer Res. (1998) pmid: 9823339
14. Pawlik TM, et al. Dis. Markers (2004) pmid: 15528785
15. Boland CR, et al. Gastroenterology (2010) pmid: 20420947
16. Samstein RM, et al. Nat. Genet. (2019) pmid: 30643254
17. Goodman AM, et al. Mol. Cancer Ther. (2017) pmid: 28835386
18. Goodman AM, et al. Cancer Immunol Res (2019) pmid: 31405947
19. Cristescu R, et al. Science (2018) pmid: 30309915
20. Ready N, et al. J. Clin. Oncol. (2019) pmid: 30785829
21. Hellmann MD, et al. N. Engl. J. Med. (2018) pmid: 29658845
22. Hellmann MD, et al. Cancer Cell (2018) pmid: 29657128
23. Hellmann MD, et al. Cancer Cell (2018) pmid: 29731394
24. Rozeman EA, et al. Nat Med (2021) pmid: 33558721
25. Sharma P, et al. Cancer Cell (2020) pmid: 32916128
26. Zhao J, et al. Nat. Med. (2019) pmid: 30742119
27. Touat M, et al. Nature (2020) pmid: 32322066
28. Bouffet E, et al. J. Clin. Oncol. (2016) pmid: 27001570
29. Johanns TM, et al. Cancer Discov (2016) pmid: 27683556
30. Lukas RV, et al. J. Neurooncol. (2018) pmid: 30073642
31. Chalmers ZR, et al. Genome Med (2017) pmid: 28420421
32. Patel RR, et al. Pediatr Blood Cancer (2020) pmid: 32386112
33. Johnson A, et al. Oncologist (2017) pmid: 28912153
34. Draaisma K, et al. Acta Neuropathol Commun (2015) pmid: 26699864
35. Wang L, et al. BMC Cancer (2020) pmid: 32164609
36. Pfeifer GP, et al. Mutat. Res. (2005) pmid: 15748635
37. Hill VK, et al. Annu Rev Genomics Hum Genet (2013) pmid: 23875803
38. Pfeifer GP, et al. Oncogene (2002) pmid: 12379884
39. Rizvi NA, et al. Science (2015) pmid: 25765070
40. Johnson BE, et al. Science (2014) pmid: 24336570
41. Choi S, et al. Neuro-oncology (2018) pmid: 29452419
42. Cancer Genome Atlas Research Network, et al. Nature (2013) pmid: 23636398
43. Briggs S, et al. J. Pathol. (2013) pmid: 23447401
44. Heitzer E, et al. Curr. Opin. Genet. Dev. (2014) pmid: 24583393
45. Nature (2012) pmid: 22810696
46. Roberts SA, et al. Nat. Rev. Cancer (2014) pmid: 25568919
47. Marabelle A, et al. Lancet Oncol. (2020) pmid: 32919526
48. Sepúlveda-Sánchez JM, et al. Neuro-oncology (2017) pmid: 28575464
49. Tanaka S, et al. Sci Rep (2019) pmid: 30644426
50. Chi AS, et al. JCO Precis Oncol (2020) pmid: 32923886
51. Reardon DA, et al. Neuro-oncology (2015) pmid: 25140039
52. Blumenthal DT, et al. J. Neurooncol. (2016) pmid: 27531351
53. Mellinghoff IK, et al. N. Engl. J. Med. (2005) pmid: 16282176
54. Franceschi E, et al. Br. J. Cancer (2007) pmid: 17353924
55. Chakravarti A, et al. Int. J. Radiat. Oncol. Biol. Phys. (2013) pmid: 23182702
56. Hegi ME, et al. Mol. Cancer Ther. (2011) pmid: 21471286
57. Jiang Z, et al. PLoS ONE (2013) pmid: 23441167
58. Licitra L, et al. Ann. Oncol. (2011) pmid: 21048039
59. Smyth EC, et al. Gut (2021) pmid: 33199443
60. Lassman et al., 2019; Neuro-Oncology Abstract ACTR-21
61. Van Den Bent M, et al. Neuro Oncol (2020) pmid: 31747009
62. Byeon S, et al. Target Oncol (2020) pmid: 32107712
63. Jonsson P, et al. Clin. Cancer Res. (2019) pmid: 31263031
64. Brennan CW, et al. Cell (2013) pmid: 24120142
65. Ceccarelli M, et al. Cell (2016) pmid: 26824661
66. Thomas AA, et al. Neuro-oncology (2017) pmid: 28472509
67. Yan et al. 2020; DOI:10.1200/PO.19.00385
68. Srividya MR, et al. J. Clin. Pathol. (2010) pmid: 20702468
69. Das P, et al. J Clin Neurosci (2011) pmid: 20888234
70. Smith JS, et al. J. Natl. Cancer Inst. (2001) pmid: 11504770
71. Ciardiello F, et al. N. Engl. J. Med. (2008) pmid: 18337605
72. Liang Z, et al. BMC Cancer (2010) pmid: 20637128
73. Bhargava R, et al. Mod. Pathol. (2005) pmid: 15920544
74. Yang YL, et al. Chin. Med. J. (2012) pmid: 22490401
75. Louis DN, et al. Neuro Oncol (2021) pmid: 34185076
76. Evans EK, et al. Sci Transl Med (2017) pmid: 29093181
77. Abbaspour Babaei M, et al. Drug Des Devel Ther (2016) pmid: 27536065
78. Ramaswamy A, et al. J Gastrointest Oncol (2016) pmid: 27563456
79. Demetri GD, et al. Lancet (2013) pmid: 23177515
80. Gotlib J, et al. N. Engl. J. Med. (2016) pmid: 27355533
81. Jawhar M, et al. Blood (2017) pmid: 28424161
82. Xu X, et al. Int J Clin Exp Pathol (2014) pmid: 25031773
83. Gotlib J, et al. Blood (2005) pmid: 15972446
84. Luo C, et al. Onco Targets Ther (2017) pmid: 29066909
85. Janku F, et al. ESMO Open (2022) pmid: 35753087
86. Si L, et al. J. Clin. Oncol. (2012) pmid: 22162580
87. Parikh SA, et al. Leuk Lymphoma (2010) pmid: 20038218
88. Wei X, et al. Oncol. Res. (2019) pmid: 30075827
89. Hodi FS, et al. J. Clin. Oncol. (2013) pmid: 23775962
90. Carvajal RD, et al. JAMA (2011) pmid: 21642685
91. Guo J, et al. J. Clin. Oncol. (2011) pmid: 21690468
92. Debiec-Rychter M, et al. Gastroenterology (2005) pmid: 15685537
93. Dematteo RP, et al. Lancet (2009) pmid: 19303137
94. Faivre S, et al. J. Clin. Oncol. (2005) pmid: 16135502
95. Hotte SJ, et al. J. Clin. Oncol. (2005) pmid: 15659505
96. Alcedo JC, et al. Head Neck (2004) pmid: 15350030
97. Brandwein JM, et al. Leukemia (2011) pmid: 21403650
98. Reardon DA, et al. Br. J. Cancer (2009) pmid: 19904263
99. Lee SJ, et al. Oncologist (2015) pmid: 26424760
100. Llovet JM, et al. Clin. Cancer Res. (2012) pmid: 22374331
101. Zhang HL, et al. Clin Genitourin Cancer (2013) pmid: 23058498
102. Seino S, et al. Gastroenterology (2014) pmid: 25450081
103. Li XF, et al. Med. Oncol. (2009) pmid: 18846437
104. Minor DR, et al. Clin. Cancer Res. (2012) pmid: 22261812
105. Mahipal A, et al. Melanoma Res. (2012) pmid: 23114504
106. Cancer Genome Atlas Research Network, et al. N. Engl. J. Med. (2015) pmid: 26061751
107. Nobusawa S, et al. Neuropathology (2011) pmid: 21382095
108. Joensuu H, et al. J. Pathol. (2005) pmid: 16021678
109. Burford A, et al. PLoS ONE (2013) pmid: 23990986
110. Holtkamp N, et al. Neuro-oncology (2007) pmid: 17504929
111. Puputti M, et al. Mol. Cancer Res. (2006) pmid: 17189383
112. Skardelly M, et al. Transl Oncol (2009) pmid: 19701495
113. Int. J. Biochem. Cell Biol. (1999) pmid: 10582339
114. Semin. Oncol. (2004) pmid: 15175998
115. Gao J, et al. Sci Signal (2013) pmid: 23550210
116. Zack TI, et al. Nat. Genet. (2013) pmid: 24071852
117. Beroukhim R, et al. Nature (2010) pmid: 20164920
118. Arefi M, et al. Int. J. Hematol. (2012) pmid: 22806436
119. Baccarani M, et al. Haematologica (2007) pmid: 17666373
120. Cassier PA, et al. Clin. Cancer Res. (2012) pmid: 22718859
121. Chalmers ZR, et al. Blood Cancer J (2015) pmid: 25658984
122. Cools J, et al. N. Engl. J. Med. (2003) pmid: 12660384
123. Curtis CE, et al. Br. J. Haematol. (2007) pmid: 17555450
124. Debiec-Rychter M, et al. Eur. J. Cancer (2004) pmid: 15010069
125. Dileo P, et al. Int. J. Cancer (2011) pmid: 20473908
126. Fanta PT, et al. J. Clin. Oncol. (2015) pmid: 24638008
127. Florian S, et al. Leuk. Res. (2006) pmid: 16406018
128. Frenard C, et al. JAAD Case Rep (2016) pmid: 27051816
129. Griffin JH, et al. Proc. Natl. Acad. Sci. U.S.A. (2003) pmid: 12808148
130. Heinrich MC, et al. J. Clin. Oncol. (2003) pmid: 14645423
131. Helbig G, et al. Br. J. Haematol. (2009) pmid: 19120352
132. Helbig G, et al. Am. J. Hematol. (2014) pmid: 24009127
133. Hus M, et al. Leuk. Res. (2011) pmid: 21093052
134. Ikezoe T, et al. Leuk. Res. (2010) pmid: 20303172
135. Intermesoli T, et al. Br. J. Haematol. (2009) pmid: 19735261
136. Jain N, et al. Leuk. Res. (2009) pmid: 19013640
137. Jovanovic JV, et al. Blood (2007) pmid: 17299092
138. Kang HJ, et al. Acta Oncol (2012) pmid: 22150077
139. Klion AD, et al. Blood (2004) pmid: 14504092
140. Kobayashi M, et al. Respirology (2009) pmid: 19192229
141. Kocáková I, et al. Klin Onkol (2014) pmid: 24635438
142. Metzgeroth G, et al. Br. J. Haematol. (2008) pmid: 18950453
143. Murayama Y, et al. World J Gastrointest Oncol (2012) pmid: 22645636
144. Ogbogu PU, et al. J. Allergy Clin. Immunol. (2009) pmid: 19910029
145. Ohnishi H, et al. Br. J. Haematol. (2006) pmid: 16856885
146. Pardanani A, et al. Blood (2003) pmid: 12842979
147. Pardanani A, et al. Blood (2004) pmid: 15284118
148. Qu SQ, et al. Oncotarget (2016) pmid: 27120808
149. Score J, et al. Leukemia (2006) pmid: 16498388

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

150. Shah S, et al. J Hematol Oncol (2014) pmid: 24669761
151. Sugimoto Y, et al. Cancer Genet (2015) pmid: 26319757
152. Volz HC, et al. Int. J. Cardiol. (2011) pmid: 20609486
153. von Bubnoff N, et al. Leukemia (2005) pmid: 15618966
154. Walz C, et al. Genes Chromosomes Cancer (2006) pmid: 16845659
155. Yoo C, et al. Cancer Res Treat (2016) pmid: 26130666
156. Al-Riyami AZ, et al. Leuk. Lymphoma (2013) pmid: 23157309
157. Lierman E, et al. Blood (2006) pmid: 16645167
158. Lierman E, et al. Leukemia (2009) pmid: 19212337
159. Metzgeroth G, et al. Leukemia (2012) pmid: 21818111
160. Roubaud G, et al. Ann. Oncol. (2012) pmid: 22294526
161. von Bubnoff N, et al. Oncogene (2011) pmid: 20972453
162. Hochhaus A, et al. J. Cancer Res. Clin. Oncol. (2013) pmid: 24057647
163. Tabouret E, et al. Leuk. Res. (2011) pmid: 20832858
164. Dewaele B, et al. Clin. Cancer Res. (2008) pmid: 18794084
165. Weisberg E, et al. Gastroenterology (2006) pmid: 17087936
166. Brohl AS, et al. Clin Sarcoma Res (2015) pmid: 26396737
167. Grellety T, et al. Future Sci OA (2015) pmid: 28031906
168. Kollàr A, et al. Clin Sarcoma Res (2014) pmid: 25905001
169. Jaku et al., 2017; ASCO Abstract 2515
170. Phillips JJ, et al. Brain Pathol. (2013) pmid: 23438035
171. Motomura K, et al. J. Neuropathol. Exp. Neurol. (2013) pmid: 23242283
172. Sottoriva A, et al. Proc. Natl. Acad. Sci. U.S.A. (2013) pmid: 23412337
173. Alentorn A, et al. Neuro-oncology (2012) pmid: 23074200
174. Song K, et al. Am J Cancer Res (2018) pmid: 29888103
175. Szerlip NJ, et al. Proc. Natl. Acad. Sci. U.S.A. (2012) pmid: 22323597
176. Andrae J, et al. Genes Dev. (2008) pmid: 18483217
177. Flavahan WA, et al. Nature (2016) pmid: 26700815
178. Roszik J, et al. Sci Rep (2016) pmid: 26787600
179. Verhaak RG, et al. Cancer Cell (2010) pmid: 20129251
180. Koschmann C, et al. Oncotarget (2016) pmid: 27582545
181. Puget S, et al. PLoS ONE (2012) pmid: 22389665
182. Williamson CT, et al. Nat Commun (2016) pmid: 27958275
183. Aggarwal et al., 2021; ESMO Abstract 512O
184. Thomas A, et al. J. Clin. Oncol. (2018) pmid: 29252124
185. Yap TA, et al. J Clin Oncol (2020) pmid: 32568634
186. Bitler BG, et al. Nat. Med. (2015) pmid: 25686104
187. Kim KH, et al. Nat. Med. (2015) pmid: 26552009
188. Papadopoulos et al., 2022; ENA Abstract 188
189. Wiegand KC, et al. BMC Cancer (2014) pmid: 24559118
190. Huang HN, et al. Mod. Pathol. (2014) pmid: 24336158
191. Samartzis EP, et al. Oncotarget (2014) pmid: 24979463
192. Okamura R, et al. J Immunother Cancer (2020) pmid: 32111729
193. Yokoyama Y, et al. J Gynecol Oncol (2014) pmid: 24459582
194. Katagiri A, et al. Mod. Pathol. (2012) pmid: 22101352
195. Xie C, et al. Tumour Biol. (2014) pmid: 24833095
196. Tate JG, et al. Nucleic Acids Res. (2019) pmid: 30371878
197. Cerami E, et al. Cancer Discov (2012) pmid: 22588877
198. Wu RC, et al. Cancer Biol. Ther. (2014) pmid: 24618703
199. Jones S, et al. Hum. Mutat. (2012) pmid: 22009941
200. Dulak AM, et al. Nat. Genet. (2013) pmid: 23525077
201. Streppel MM, et al. Oncogene (2014) pmid: 23318448
202. Jiao Y, et al. J. Pathol. (2014) pmid: 24293293
203. Ross JS, et al. Oncologist (2014) pmid: 24563076
204. Huang HN, et al. Histopathology (2015) pmid: 25195947
205. Hussein YR, et al. Mod. Pathol. (2015) pmid: 25394778
206. Bosse T, et al. Mod. Pathol. (2013) pmid: 23702729
207. Allo G, et al. Mod. Pathol. (2014) pmid: 23887303
208. Chou A, et al. Hum. Pathol. (2014) pmid: 24925223
209. Ye J, et al. Hum. Pathol. (2014) pmid: 25311944
210. Wei XL, et al. World J. Gastroenterol. (2014) pmid: 25561809
211. Chen K, et al. Proc. Natl. Acad. Sci. U.S.A. (2015) pmid: 25583476
212. Wang K, et al. Nat. Genet. (2011) pmid: 22037554
213. Abe H, et al. Virchows Arch. (2012) pmid: 22915242
214. Wang DD, et al. PLoS ONE (2012) pmid: 22808142
215. Wiegand KC, et al. Hum. Pathol. (2014) pmid: 24767857
216. Katagiri A, et al. Int. J. Gynecol. Cancer (2012) pmid: 22274316
217. Cho H, et al. Hum. Pathol. (2013) pmid: 23427874
218. Gui Y, et al. Nat. Genet. (2011) pmid: 21822268
219. Balbás-Martínez C, et al. PLoS ONE (2013) pmid: 23650517
220. Faraj SF, et al. Hum. Pathol. (2014) pmid: 25175170
221. Rahman M, et al. Hum. Pathol. (2013) pmid: 22939958
222. Maeda D, et al. Int J Mol Sci (2010) pmid: 21614196
223. Lowery WJ, et al. Int. J. Gynecol. Cancer (2012) pmid: 22193641
224. Fadare O, et al. Mod. Pathol. (2013) pmid: 23524907
225. Mao TL, et al. Am. J. Surg. Pathol. (2013) pmid: 24076775
226. Zhang X, et al. Cancer Epidemiol (2012) pmid: 21889920
227. Mamo A, et al. Oncogene (2012) pmid: 21892209
228. Zhao J, et al. Tumour Biol. (2014) pmid: 24430365
229. Lichner Z, et al. Am. J. Pathol. (2013) pmid: 23416164
230. Feng F, et al. Int J Clin Oncol (2021) pmid: 33387086
231. Conci S, et al. Updates Surg (2020) pmid: 32020551
232. Simbolo M, et al. Sci Rep (2018) pmid: 29740198
233. Ruzzenente A, et al. Ann. Surg. Oncol. (2016) pmid: 26717940
234. Guan B, et al. Cancer Res. (2011) pmid: 21900401
235. Wiegand KC, et al. N. Engl. J. Med. (2010) pmid: 20942669
236. Jones S, et al. Science (2010) pmid: 20826764
237. Yan HB, et al. Carcinogenesis (2014) pmid: 24293408
238. Huang J, et al. Nat. Genet. (2012) pmid: 22922871
239. Chan-On W, et al. Nat. Genet. (2013) pmid: 24185513
240. Zang ZJ, et al. Nat. Genet. (2012) pmid: 22484628
241. Dickson MA, et al. J. Clin. Oncol. (2013) pmid: 23569312
242. Flaherty KT, et al. Clin. Cancer Res. (2012) pmid: 22090362
243. Patnaik A, et al. Cancer Discov (2016) pmid: 27217383
244. Infante JR, et al. Clin. Cancer Res. (2016) pmid: 27542767
245. Dickson et al., 2019; ASCO Abstract 11004
246. Dickson MA, et al. JAMA Oncol (2016) pmid: 27124835
247. Peguero et al., 2016; ASCO Abstract 2528
248. Zheng S, et al. Genes Dev. (2013) pmid: 23796897
249. Kim H, et al. Proc. Natl. Acad. Sci. U.S.A. (2010) pmid: 20080666
250. Ruano Y, et al. Am. J. Clin. Pathol. (2009) pmid: 19141386
251. Fischer U, et al. Mol. Cancer Res. (2008) pmid: 18403636
252. Bäcklund LM, et al. Br. J. Cancer (2005) pmid: 15970925
253. Choi YJ, et al. Oncogene (2014) pmid: 23644662
254. Cell (1995) pmid: 7736585
255. Musgrove EA, et al. Nat. Rev. Cancer (2011) pmid: 21734724
256. Wikman H, et al. Genes Chromosomes Cancer (2005) pmid: 15543620
257. Rao SK, et al. J. Neurooncol. (2010) pmid: 19609742
258. Chung L, et al. Am. J. Surg. Pathol. (2009) pmid: 19574885
259. Ragazzini P, et al. Histol. Histopathol. (2004) pmid: 15024701
260. Dujardin F, et al. Mod. Pathol. (2011) pmid: 21336260
261. Zhang K, et al. Cancer Res. (2013) pmid: 23393200
262. Horvai AE, et al. Mod. Pathol. (2009) pmid: 19734852
263. Cheok CF, et al. Nat Rev Clin Oncol (2011) pmid: 20975744
264. Ohnstad HO, et al. Cancer (2013) pmid: 23165797
265. Gamble LD, et al. Oncogene (2012) pmid: 21725357
266. Zhang et al., 2019; ASCO Abstract 3124
267. Rasco et al., 2019; ASCO Abstract 3126
268. Tolcher et al., 2021; ASCO Abstract 2506
269. Martinelli et al., 2016; EHA21 Abstract S504
270. Daver et al., 2018; ASH Abstract 767
271. Mascarenhas et al., 2019; ASH Abstract 134
272. Shustov et al., 2018; ASH Abstract 1623
273. Sallman et al., 2018; ASH Abstract 4066
274. Meric-Bernstam et al., 2017; ASCO Abstract 2505
275. Fischer U, et al. Int. J. Cancer (2010) pmid: 19839052
276. Korkolopoulou P, et al. Br J Cancer (1997) pmid: 9155045
277. Ranuncolo SM, et al. J. Neurooncol. (2004) pmid: 15218947
278. Sdek P, et al. Mol. Cell (2005) pmid: 16337594
279. Brady M, et al. Mol. Cell. Biol. (2005) pmid: 15632057
280. Li M, et al. Mol. Cell (2004) pmid: 15053880
281. Brown CJ, et al. Nat. Rev. Cancer (2009) pmid: 19935675
282. Cordon-Cardo C, et al. Cancer Res. (1994) pmid: 8306343
283. Kato S, et al. Clin. Cancer Res. (2017) pmid: 28351930
284. Singavi et al., 2017; ESMO Abstract 1140PD
285. Rizvi H, et al. J. Clin. Oncol. (2018) pmid: 29337640
286. Garralda et al., 2022; ENA Abstract 7
287. Horiuchi D, et al. J. Exp. Med. (2012) pmid: 22430491
288. Goga A, et al. Nat. Med. (2007) pmid: 17589519
289. Molenaar JJ, et al. Proc. Natl. Acad. Sci. U.S.A. (2009) pmid: 19525400
290. Dammert MA, et al. Nat Commun (2019) pmid: 31375684
291. Mollaoglu G, et al. Cancer Cell (2017) pmid: 28089889
292. Cardnell RJ, et al. Oncotarget (2017) pmid: 29088717
293. Wang L, et al. Mol Oncol (2017) pmid: 28417568
294. Takahashi Y, et al. Ann. Oncol. (2015) pmid: 25632068
295. Li Y, et al. Thyroid (2018) pmid: 30226440
296. Mahadevan D, et al. PLoS ONE (2014) pmid: 24893165
297. Park SI, et al. Target Oncol (2019) pmid: 31429028
298. Helfrich BA, et al. Mol. Cancer Ther. (2016) pmid: 27496133
299. Hook KE, et al. Mol. Cancer Ther. (2012) pmid: 22222631
300. Yang D, et al. Proc. Natl. Acad. Sci. U.S.A. (2010) pmid: 20643922
301. He J, et al. Anticancer Drugs (2019) pmid: 30540594
302. Shroff EH, et al. Proc. Natl. Acad. Sci. U.S.A. (2015) pmid: 25964345
303. Effenberger M, et al. Oncotarget (2017) pmid: 29156762
304. Qu X, et al. Biochem. Biophys. Res. Commun. (2018) pmid: 30103944
305. Xiang Y, et al. J. Clin. Invest. (2015) pmid: 25915584
306. Delmore JE, et al. Cell (2011) pmid: 21889194

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

307. Bandopadhayay P, et al. Clin. Cancer Res. (2014) pmid: 24297863
308. Lovén J, et al. Cell (2013) pmid: 23582323
309. Otto C, et al. Neoplasia (2019) pmid: 31734632
310. Dong LH, et al. J Hematol Oncol (2013) pmid: 23866964
311. Pei Y, et al. Cancer Cell (2016) pmid: 26977882
312. Fu XH, et al. Acta Pharmacol. Sin. (2019) pmid: 30224636
313. Owonikoko TK, et al. J Thorac Oncol (2020) pmid: 31655296
314. Ganesan P, et al. Mol. Cancer Ther. (2014) pmid: 25253784
315. Pereira CB, et al. PLoS ONE (2013) pmid: 23555992
316. Yasojima H, et al. Eur. J. Cancer (2011) pmid: 21741827
317. Arango D, et al. Cancer Res. (2001) pmid: 11406570
318. Bottone MG, et al. Exp. Cell Res. (2003) pmid: 14516787
319. Felicella MM, et al. Clin. Neuropathol. () pmid: 22720694
320. Faria et al. 2008; 18369647
321. Hu YH, et al. Chin. Med. J. (2012) pmid: 22884072
322. De Salvo M, et al. Int. J. Radiat. Biol. (2011) pmid: 21405945
323. Cenci T, et al. Am. J. Clin. Pathol. (2012) pmid: 22912356
324. Lassman AB, et al. Neuron Glia Biol. (2004) pmid: 17047730
325. Dang CV, et al. Semin. Cancer Biol. (2006) pmid: 16904903
326. Nesbit CE, et al. Oncogene (1999) pmid: 10378696
327. Blancato J, et al. Br. J. Cancer (2004) pmid: 15083194
328. Fromont G, et al. Hum. Pathol. (2013) pmid: 23574779
329. Matulonis U, et al. Gynecol. Oncol. (2015) pmid: 25528496
330. Pitz MW, et al. Neuro-oncology (2015) pmid: 25605819
331. Thorpe LM, et al. Proc. Natl. Acad. Sci. U.S.A. (2017) pmid: 28630349
332. Sun M, et al. Proc. Natl. Acad. Sci. U.S.A. (2010) pmid: 20713702
333. Juric et al., 2016; SABCS Abstract P3-14-01
334. Basho RK, et al. JAMA Oncol (2017) pmid: 27893038
335. Myers AP, et al. Gynecol. Oncol. (2016) pmid: 27016228
336. Day TA, et al. Clin. Cancer Res. (2019) pmid: 30420444
337. Ou O, et al. Cancer Lett. (2014) pmid: 25193464
338. Li X, et al. Nat Commun (2019) pmid: 30755611
339. Quayle SN, et al. PLoS ONE (2012) pmid: 23166678
340. Cheung LW, et al. Cancer Cell (2014) pmid: 25284480
341. Ye K, et al. Nat. Med. (2016) pmid: 26657142
342. Cheung LW, et al. Cancer Discov (2011) pmid: 21984976
343. Urick ME, et al. Cancer Res. (2011) pmid: 21478295
344. Munkley J, et al. Oncoscience (2015) pmid: 26501081
345. Cizkova M, et al. BMC Cancer (2013) pmid: 24229379
346. Qian ZR, et al. J. Clin. Oncol. (2013) pmid: 23980085
347. Huang CH, et al. Cell Cycle (2008) pmid: 18418043
348. Taniguchi CM, et al. Cancer Res. (2010) pmid: 20530665
349. Luo J, et al. Cell Metab. (2006) pmid: 16679293
350. Ueki K, et al. J. Biol. Chem. (2003) pmid: 14504291
351. Mauvais-Jarvis F, et al. J. Clin. Invest. (2002) pmid: 11781359
352. Luo J, et al. Proc. Natl. Acad. Sci. U.S.A. (2005) pmid: 16006513
353. Jaiswal BS, et al. Cancer Cell (2009) pmid: 19962665
354. Ko HR, et al. Cell Death Dis (2014) pmid: 24651434
355. Wu H, et al. Proc. Natl. Acad. Sci. U.S.A. (2009) pmid: 19915146
356. Huang CH, et al. Science (2007) pmid: 18079394
357. Bousquet C, et al. EMBO J. (2006) pmid: 16917505
358. Oliver MD, et al. Biosci. Rep. (2017) pmid: 28143957
359. Philp AJ, et al. Cancer Res. (2001) pmid: 11606375
360. Lucas CL, et al. J. Exp. Med. (2014) pmid: 25488983
361. Chen L, et al. Nat Commun (2018) pmid: 29636477
362. Li X, et al. Proc Natl Acad Sci U S A (2021) pmid: 34507989
363. Chen XJ, et al. Front Immunol (2021) pmid: 33953726
364. Liu D, et al. Cancers (Basel) (2022) pmid: 35681795
365. Mersman DP, et al. Genes Dev. (2009) pmid: 19346402
366. Kim TD, et al. Biochem. Biophys. Res. Commun. (2008) pmid: 18078810
367. Abidi FE, et al. J. Med. Genet. (2008) pmid: 18697827
368. Debeb BG, et al. Breast Cancer Res. Treat. (2012) pmid: 22547109
369. Fouladi M, et al. J. Clin. Oncol. (2011) pmid: 21825264
370. Groth C, et al. Semin. Cell Dev. Biol. (2012) pmid: 22309842
371. Kamstrup MR, et al. Blood (2010) pmid: 20538790
372. Kridel R, et al. Blood (2012) pmid: 22210878
373. Krop I, et al. J. Clin. Oncol. (2012) pmid: 22547604
374. Rosati E, et al. Int. J. Cancer (2013) pmid: 23001755
375. Samon JB, et al. Mol. Cancer Ther. (2012) pmid: 22504949
376. Lehal R, et al. Proc Natl Acad Sci U S A (2020) pmid: 32601208
377. Ferrarotto et al., 2020; ESMO Abstract 919MO
378. Knoechel B, et al. Cold Spring Harb Mol Case Stud (2015) pmid: 27148573
379. Lopez-Miranda et al., 2021; ASCO Abstract 3020
380. Nature (2008) pmid: 18772890
381. Yip S, et al. J. Pathol. (2012) pmid: 22072542
382. Xu P, et al. J. Neurooncol. (2010) pmid: 19771395
383. Xu P, et al. Pathol. Oncol. Res. (2009) pmid: 19424825
384. Brandt WD, et al. J. Neuropathol. Exp. Neurol. (2015) pmid: 25575134
385. Jiang L, et al. J Clin Neurosci (2011) pmid: 21251836
386. Wang NJ, et al. Proc. Natl. Acad. Sci. U.S.A. (2011) pmid: 22006338
387. Klinakis A, et al. Nature (2011) pmid: 21562564
388. Penton AL, et al. Semin. Cell Dev. Biol. (2012) pmid: 22306179
389. Kopan R, et al. Cell (2009) pmid: 19379690
390. Andrawes MB, et al. J. Biol. Chem. (2013) pmid: 23839946
391. Rebay I, et al. Cell (1991) pmid: 1657403
392. Ge C, et al. BMC Dev. Biol. (2008) pmid: 18445292
393. Aster JC, et al. Mol. Cell. Biol. (2000) pmid: 11003647
394. Weng AP, et al. Science (2004) pmid: 15472075
395. Deregowski V, et al. J. Bone Miner. Res. (2006) pmid: 16869730
396. Uchibori M, et al. Oncol. Rep. (2017) pmid: 28791383
397. Liu J, et al. Proc Natl Acad Sci U S A (2013) pmid: 24277854
398. Nat Rev Clin Oncol (2017) pmid: 27245281
399. Duperret EK, et al. Mol Ther (2018) pmid: 29249395
400. Chiappori AA, et al. Ann Oncol (2015) pmid: 25467017
401. Killela PJ, et al. Proc. Natl. Acad. Sci. U.S.A. (2013) pmid: 23530248
402. Killela PJ, et al. Oncotarget (2014) pmid: 24722048
403. Nonoguchi N, et al. Acta Neuropathol. (2013) pmid: 23955565
404. Liu X, et al. Cell Cycle (2013) pmid: 23603989
405. Koelsche C, et al. Acta Neuropathol. (2013) pmid: 24154961
406. Arita H, et al. Acta Neuropathol. (2013) pmid: 23764841
407. Reitman ZJ, et al. Acta Neuropathol. (2013) pmid: 24217890
408. Shay JW, et al. Semin. Cancer Biol. (2011) pmid: 22015685
409. Shay JW, et al. Eur. J. Cancer (1997) pmid: 9282118
410. Kim NW, et al. Science (1994) pmid: 7605428
411. Hanahan D, et al. Cell (2000) pmid: 10647931
412. Horn S, et al. Science (2013) pmid: 23348503
413. Huang FW, et al. Science (2013) pmid: 23348506
414. Vinagre J, et al. Nat Commun (2013) pmid: 23887589
415. Weller M, et al. Nat Rev Clin Oncol (2021) pmid: 33293629
416. Hasselbalch B, et al. Neuro-oncology (2010) pmid: 20406901
417. Lv S, et al. Int. J. Oncol. (2012) pmid: 22752145
418. Kulason KO, et al. J Exp Ther Oncol (2018) pmid: 29790314
419. Debiec-Rychter M, et al. Eur. J. Cancer (2006) pmid: 16624552
420. Kamenz T, et al. World J. Gastroenterol. (2006) pmid: 16570351
421. Wang YY, et al. Proc. Natl. Acad. Sci. U.S.A. (2005) pmid: 15650049
422. Hassler MR, et al. Springerplus (2014) pmid: 25674429
423. Razis E, et al. Clin. Cancer Res. (2009) pmid: 19789313
424. Reardon DA, et al. Cancer (2012) pmid: 22371319
425. Carvajal RD, et al. Clin. Cancer Res. (2015) pmid: 25695690
426. Hochhaus A, et al. J. Cancer Res. Clin. Oncol. (2015) pmid: 26002753
427. Blay JY, et al. Lancet Oncol. (2015) pmid: 25882987
428. Kajimoto N, et al. Int J Clin Exp Pathol (2015) pmid: 26722383
429. Sako H, et al. PLoS ONE (2014) pmid: 25221952
430. Hughes TP, et al. Blood (2014) pmid: 24335106
431. Takahashi N, et al. Biomark Res (2014) pmid: 24650752
432. Reichardt P, et al. Ann. Oncol. (2012) pmid: 22357255
433. Cauchi C, et al. Cancer Chemother. Pharmacol. (2012) pmid: 22119758
434. Whittle JR, et al. J Clin Neurosci (2015) pmid: 26279503
435. Douillard JY, et al. Ann. Oncol. (2014) pmid: 24718886
436. Price TJ, et al. Lancet Oncol. (2014) pmid: 24739896
437. Van Cutsem E, et al. J. Clin. Oncol. (2007) pmid: 17470858
438. Vermorken JB, et al. Lancet Oncol (2013) pmid: 23746666
439. Wirth LJ, et al. Ann. Oncol. (2010) pmid: 19892746
440. Siano M, et al. Oncologist (2017) pmid: 28592616
441. Mesía R, et al. Lancet Oncol (2015) pmid: 25596660
442. Giralt J, et al. Lancet Oncol (2015) pmid: 25596659
443. Siu LL, et al. JAMA Oncol (2016) pmid: 27930762
444. Waddell T, et al. Lancet Oncol. (2013) pmid: 23594787
445. Crawford J, et al. J Thorac Oncol (2013) pmid: 24389433
446. Schuette W, et al. Clin Lung Cancer (2015) pmid: 26094080
447. Leone F, et al. Cancer (2016) pmid: 26540314
448. Vogel A, et al. Eur J Cancer (2018) pmid: 29413685
449. Rowinsky EK, et al. J. Clin. Oncol. (2004) pmid: 15210739
450. Quintás-Cardama A, et al. Nat Clin Pract Oncol (2008) pmid: 18936790
451. Bisagni G, et al. J Thorac Oncol (2009) pmid: 19461405
452. Handolias D, et al. Br. J. Cancer (2010) pmid: 20372153
453. Dişel U, et al. Lung Cancer (2011) pmid: 20970876
454. Park SH, et al. Invest New Drugs (2012) pmid: 22270258
455. Catania C, et al. Onco Targets Ther (2014) pmid: 24855380

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.

456. Guo T, et al. Clin. Cancer Res. (2007) pmid: 17699867
457. Hu S, et al. Mol. Cancer Ther. (2008) pmid: 18483300
458. Zustovich et al., 2013; 23898124; Reardon et al.
459. Lee EQ, et al. Neuro-oncology (2012) pmid: 23099651
460. Peereboom DM, et al. Neuro-oncology (2013) pmid: 23328813
461. Hottinger AF, et al. Br. J. Cancer (2014) pmid: 24786603
462. Karajannis MA, et al. Neuro-oncology (2014) pmid: 24803676
463. Heinrich MC, et al. J. Clin. Oncol. (2008) pmid: 18955458
464. Buchbinder EI, et al. Cancer (2015) pmid: 26264378
465. Reichardt P, et al. BMC Cancer (2016) pmid: 26772734
466. Hirai F, et al. Mol Clin Oncol (2016) pmid: 27073655
467. Goemans BF, et al. Leuk. Res. (2010) pmid: 20435347
468. Pan E, et al. J. Neurooncol. (2012) pmid: 22832897
469. Kreisl TN, et al. J. Neurooncol. (2013) pmid: 23086433
470. Balaña C, et al. Target Oncol (2014) pmid: 24424564
471. Hutterer M, et al. Neuro-oncology (2014) pmid: 24311637

Disclaimer: Foundation Medicine Inc. only provides PDF report as an official issuance of the test result. Any other transformed format is not an "official / formal solution" and not guarantee the accuracy of this conversion. It is suggested the hospital to verify the outputs and validate the suitability of use.

© 2023 Foundation Medicine, Inc. All rights reserved.